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ELECTROSTATIC DISCHARGE CIRCUIT AND METHOD OF OPERATING SAME

Abstract

An electrostatic discharge (ESD) circuit includes a first ESD detection circuit between a first node having a first voltage and a second node having a second voltage, a first clamp circuit including a first transistor between the first, second and a third node, a first ESD assist circuit between the second and third node, and configured to clamp a third voltage of the third node at the second voltage during an ESD event at the first or second node, a second ESD detection circuit between the second and a fourth node, a second clamp circuit including a second transistor between the second, the node and a fifth node, and a second ESD assist circuit between the fourth and the fifth node, and configured to clamp a fourth voltage of the fifth node at the second voltage during an ESD event at the second or the fourth node.

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Background/Summary

PRIORITY CLAIM [0001] The present application is a continuation of U.S. application Ser. No. 18/545,851, filed Dec. 19, 2023, which is a continuation of U.S. application Ser. No. 17/452,281, filed Oct. 26, 2021, now U.S. Pat. No. 11,848,554, issued Dec. 19, 2023, which claims the benefit of U.S. Provisional Application No. 63/177,629, filed Apr. 21, 2021, which are herein incorporated by reference in their entireties.

BACKGROUND

[0002] The recent trend in miniaturizing integrated circuits (ICs) has resulted in smaller devices which consume less power, yet provide more functionality at higher speeds than before. The miniaturization process has also increased the devices' susceptibility to electrostatic discharge (ESD) events due to various factors, such as thinner dielectric thicknesses and associated lowered dielectric breakdown voltages. ESD is one of the causes of electronic circuit damage and is also one of the considerations in semiconductor advanced technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a schematic block diagram of an integrated circuit, in accordance with some embodiments.

[0005] FIGS. 2A-2E are corresponding circuit diagrams of corresponding integrated circuits, in accordance with some embodiments.

[0006] FIGS. 3A-3E are corresponding cross-sectional views of corresponding integrated circuits, in accordance with some embodiments.

[0007] FIGS. 4A-4E are corresponding circuit diagrams of corresponding integrated circuits, in accordance with some embodiments.

[0008] FIGS. 5A-5E are corresponding cross-sectional views of corresponding integrated circuits, in accordance with some embodiments.

[0009] FIGS. 6A-6B are corresponding circuit diagrams of corresponding integrated circuits, in accordance with some embodiments.

[0010] FIGS. 7A-7B are corresponding circuit diagrams of corresponding integrated circuits, in accordance with some embodiments.

[0011] FIGS. 8A-8B are top-views of corresponding integrated circuits, in accordance with some embodiments.

[0012] FIG. **9** is a flowchart of a method of operating an ESD circuit, in accordance with some embodiments.

[0013] FIG. **10** is a flow chart of a method of manufacturing an integrated circuit, in accordance with some embodiments.

DETAILED DESCRIPTION

[0014] The following disclosure provides different embodiments, or examples, for implementing features of the provided subject matter. Specific examples of components, materials, values, steps, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not limiting. Other components, materials, values, steps, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0015] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0016] In some embodiments, an electrostatic discharge (ESD) circuit includes an ESD detection circuit coupled between a first node and a second node. In some embodiments, the first node has a first voltage, and the second node has a second voltage.

[0017] In some embodiments, the ESD circuit further includes a clamp circuit that includes a first transistor of a first type. The first transistor has a first gate, a first drain, a first source and a first body terminal. The first gate is coupled to at least the ESD detection circuit by a third node. The first drain is coupled to the second node. The first source and the first body terminal are coupled together at the first node.

[0018] The ESD circuit further includes an ESD assist circuit that is coupled between the first node and the third node. During an ESD event at the first node or the second node of the present disclosure, the ESD assist circuit is configured to clamp a third voltage of the third node at the first voltage thereby controlling a gate potential of the first transistor and causing the first transistor to turn-on. In response to turning on, the first transistor is configured to discharge an ESD current of the ESD event in a forward ESD direction from the second node to the first node, in accordance with some embodiments.

[0019] In some embodiments, by including the ESD assist circuit in the integrated circuit of the present disclosure, the ESD assist circuit is configured to reduce capacitive coupling effects between the first node and the second node and effectively lowers the turn-on resistance of the first transistor. In some embodiments, by lowering the turn-on resistance of the first transistor causing the first transistor to turn-on faster and stronger than other approaches results in improved ESD performance and improved ESD robustness compared to other approaches.

[0020] FIG. **1** is a schematic block diagram of an integrated circuit **100**, in accordance with some embodiments.

[0021] Integrated circuit **100** comprises an internal circuit **102**, a voltage supply node **104**, a reference voltage supply node **106**, an input/output (IO) pad **108**, a diode **110**, a diode **112** and an ESD clamp **120**. In some embodiments, at least integrated circuit **100**, **200A-200E** (FIGS. 2A-2E),

300A-300E (FIGS. 3A-3E), **400A-400E** (FIGS. 4A-4E), **500A-500E** (FIGS. 5A-5E), **600A-600B** (FIGS. 6A-6B), **700A-700B** (FIGS. 7A-7B) or **800A-800B** (FIGS. 8A-8B) is incorporated on a single integrated circuit (IC), or on a single semiconductor substrate. In some embodiments, at least integrated circuit **100**, **200A-200E** (FIGS. 2A-2E), **300A-300E** (FIGS. 3A-3E), **400A-400E** (FIGS. 4A-4E), **500A-500E** (FIGS. 5A-5E), **600A-600B** (FIGS. 6A-6B), **700A-700B** (FIGS. 7A-7B) or **800A-800B** (FIGS. 8A-8B) includes one or more ICs incorporated on one or more single semiconductor substrate.

[0022] Internal circuit **102** is coupled to the IO pad **108**, diode **110** and diode **112**. Internal circuit **102** is configured to receive an IO signal from IO pad **108**. In some embodiments, internal circuit **102** is coupled to voltage supply node **104** (e.g., VDD) and reference voltage supply node **106** (e.g., VSS). In some embodiments, internal circuit **102** is configured to receive a supply voltage VDD from voltage supply node **104** (e.g., VDD), and a reference voltage VSS from reference voltage supply node **106** (e.g., VSS).

[0023] Internal circuit **102** includes circuitry configured to generate or process the IO signal received by or output to IO pad **108**. In some embodiments, internal circuit **102** comprises core circuitry configured to operate at a voltage lower than supply voltage VDD of voltage supply node **104**. In some embodiments, internal circuit **102** includes at least one n-type or p-type transistor device. In some embodiments, internal circuit **102** includes at least a logic gate cell. In some embodiments, a logic gate cell includes an AND, OR, NAND, NOR, XOR, INV, AND-OR-Invert (AOI), OR-AND-Invert (OAI), MUX, Flip-flop, BUFF, Latch, delay, or clock cells. In some embodiments, internal circuit **102** includes at least a memory cell. In some embodiments, the memory cell includes a static random access memory (SRAM), a dynamic RAM (DRAM), a resistive RAM (RRAM), a magnetoresistive RAM (MRAM) or read only memory (ROM). In some embodiments, internal circuit **102** includes one or more active or passive elements. Examples of active elements include, but are not limited to, transistors and diodes. Examples of transistors include, but are not limited to, metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), or the like), FinFETs, and planar MOS transistors with raised source/drain. Examples of passive elements include, but are not limited to, capacitors, inductors, fuses, and resistors.

[0024] Voltage supply node **104** is coupled to diode **110** and ESD clamp **120**. Reference voltage supply node **106** is coupled to diode **112** and ESD clamp **120**. Voltage supply node **104** is configured to receive supply voltage VDD for normal operation of internal circuit **102**. Similarly, reference voltage supply node **106** is configured to receive reference supply voltage VSS for normal operation of internal circuit **102**. In some embodiments, at least voltage supply node **104** is a voltage supply pad. In some embodiments, at least reference voltage supply node **106** is a reference voltage supply pad. In some embodiments, a pad is at least a conductive surface, a pin, a node or a bus. Voltage supply node **104** or reference voltage supply node **106** is also referred to as a power supply voltage bus or rail. In the example configuration in FIG. 1, 2A-2E, 3A-3E, 4A-4E, 5A-5E, 6A-6B, 7A-7B or 8A-8B supply voltage VDD is a positive supply voltage, voltage supply node **104** is a positive power supply voltage, reference supply voltage VSS is a ground supply voltage, and reference voltage supply node **106** is a ground voltage terminal. Other power supply arrangements are within the scope of the present disclosure.

[0025] IO pad **108** is coupled to internal circuit **102**. IO pad **108** is configured to receive IO signal from internal circuit **102** or configured to output IO signal to internal circuit **102**. IO pad **108** is at least a pin that is coupled to internal circuit **102**. In some embodiments, IO pad **108** is a node, a bus or a conductive surface that is coupled to internal circuit **102**.

[0026] Diode **110** is coupled between voltage supply node **104** and IO pad **108**. An anode of diode **110** is coupled to internal circuit **102**, IO pad **108** and a cathode of diode **112**. A cathode of diode

110 is coupled to voltage supply node **104** and ESD clamp **120**. In some embodiments, diode **110** is a pull-up diode or referred to as a p+ diode. For example, in these embodiments, the p+-diode is formed between a p-well region (not shown) and an n-well region (not shown), and the n-well region is connected to VDD.

[0027] Diode **112** is coupled between reference voltage supply node **106** and IO pad **108**. An anode of diode **112** is coupled to reference voltage supply node **106** and ESD clamp **120**. A cathode of diode **112** is coupled to internal circuit **102**, IO pad **108** and the anode of diode **110**. In some embodiments, diode **112** is a pull-down diode or referred to as an n+ diode. For example, in these embodiments, the n+-diode is formed between an n+ junction (not shown) and a p-substrate (not shown), and the P-substrate is connected to ground or VSS.

[0028] Diodes **110** and **112** are configured to have a minimal impact on the normal behavior (e.g., no ESD conditions or events) of internal circuit **102** or integrated circuit **100**. In some embodiments, an ESD event occurs when an ESD voltage or current higher than a level of voltage or current expected during the normal operation of internal circuit **102** is applied to at least voltage supply node **104**, reference voltage supply node **106** or IO pad **108**.

[0029] When no ESD events occur, diodes **110** and **112** do not affect the operation of integrated circuit **100**. During an ESD event, diode **110** is configured to transfer voltage or current between voltage supply node **104** and IO pad **108** dependent upon whether diode **110** is forward biased or reverse biased, and the voltage levels of the voltage supply node **104** and IO pad **108**.

[0030] For example, during a Positive-to-VDD (PD) mode of ESD stress or event, diode **110** is forward biased and is configured to transfer voltage or current from IO pad **108** to voltage supply node **104**. In PD-mode, a positive ESD stress or ESD voltage (at least greater than supply voltage VDD) is applied to IO pad **108**, while voltage supply node **104** (e.g., VDD) is ground and reference voltage supply node **106** (e.g., VSS) is floating.

[0031] For example, during a Negative-to-VDD (ND) mode of ESD stress or event, diode **110** is reverse biased and is configured to transfer voltage or current from voltage supply node **104** to IO pad **108**. In ND-mode, a negative ESD stress is received by IO pad **108**, while the voltage supply node **104** (e.g., VDD) is ground and reference voltage supply node **106** (e.g., VSS) is floating.

[0032] During an ESD event, diode **112** is configured to transfer voltage or current between reference voltage supply node **106** and IO pad **108** dependent upon whether diode **112** is forward biased or reverse biased, and the voltage levels of the reference voltage supply node **106** and IO pad **108**.

[0033] For example, during a Positive-to-VSS (PS) mode of ESD stress or event, diode **112** is reverse biased and is configured to transfer voltage or current from IO pad **108** to reference voltage supply node **106**. In PS-mode, a positive ESD stress or ESD voltage (at least greater than reference supply voltage VSS) is applied to IO pad **108**, while voltage supply node **104** (e.g., VDD) is floating and reference voltage supply node **106** (e.g., VSS) is ground.

[0034] For example, during a Negative-to-VSS (NS) mode of ESD stress or event, diode **112** is forward biased and is configured to transfer voltage or current from reference voltage supply node **106** to IO pad **108**. In NS-mode, a negative ESD stress is received by IO pad **108**, while the voltage supply node **104** (e.g., VDD) is floating and reference voltage supply node **106** (e.g., VSS) is ground.

[0035] Other types of diodes, configurations and arrangements of at least diode **110** or **112** are within the scope of the present disclosure.

[0036] ESD clamp **120** is coupled between voltage supply node **104** (e.g., supply voltage VDD) and reference voltage supply node **106** (e.g., VSS). When no ESD event occurs, ESD clamp **120** is turned off. For example, when no ESD event occurs, ESD clamp **120** is turned off, and is therefore a nonconductive device or circuit during the normal operation of internal circuit **102**. In other words, ESD clamp **120** is turned off or is non-conductive in the absence of an ESD event.

[0037] If an ESD event occurs, ESD clamp **120** is configured to sense the ESD event, and is

configured to turn on and provide a current shunt path between voltage supply node **104** (e.g., supply voltage VDD) and reference voltage supply node **106** (e.g., VSS) to thereby discharge the ESD current. For example, when an ESD event occurs, the voltage difference across the ESD clamp **120** is equal to or greater than a threshold voltage of ESD clamp **120**, and ESD clamp **120** is turned ON thereby conducting current between voltage supply node **104** (e.g., VDD) and reference voltage supply node **106** (e.g., VSS).

[0038] During an ESD event, ESD clamp **120** is configured to turn on and discharge an ESD current (I1 or I2) in a forward ESD direction (e.g., current I1) or a reverse ESD direction (e.g., current I2). The forward ESD direction (e.g., current I1) is from the reference voltage supply node **106** (e.g., VSS) to the voltage supply node **104** (e.g., VDD). The reverse ESD direction (e.g., current I2) is from voltage supply node **104** (e.g., VDD) to reference voltage supply node **106** (e.g., VSS).

[0039] During a positive ESD surge on reference voltage supply node **106**, ESD clamp **120** is configured to turn on and discharge the ESD current I1 in a forward ESD direction from the reference voltage supply node **106** (e.g., VSS) to the voltage supply node **104** (e.g., VDD). In some embodiments, ESD clamp **120** is configured to turn on, after a PS mode (described above) of ESD, and discharge the ESD current I1 in the forward ESD direction from the reference voltage supply node **106** (e.g., VSS) to the voltage supply node **104** (e.g., VDD).

[0040] During a positive ESD surge on voltage supply node **104**, ESD clamp **120** is configured to turn on and discharge the ESD current I2 in a reverse ESD direction from voltage supply node **104** (e.g., VDD) to reference voltage supply node **106** (e.g., VSS). In some embodiments, ESD clamp **120** is configured to turn on, after a PD mode (described above) of ESD, and discharge the ESD current I2 in the reverse ESD direction from voltage supply node **104** (e.g., VDD) to reference voltage supply node **106** (e.g., VSS).

[0041] In some embodiments, ESD clamp **120** is a transient clamp. For example, in some embodiments, ESD clamp **120** is configured to handle transient or rapid ESD events, e.g., rapid changes in voltage and/or current from the ESD event. During the transient or rapid ESD, the ESD clamp **120** is configured to turn on very quickly to provide a shunt path between voltage supply node **104** (e.g., supply voltage VDD) and reference voltage supply node **106** (e.g., VSS) before the ESD event can cause damage to one or more elements within integrated circuit **100**. In some embodiments, ESD clamp **120** is configured to turn off slower than it turns on.

[0042] In some embodiments, ESD clamp **120** is a static clamp. In some embodiments, static clamps are configured to provide a static or steady-state voltage and current response. For example, static clamps are turned-on by a fixed voltage level.

[0043] In some embodiments, ESD clamp **120** includes a large NMOS transistor or PMOS transistor configured to carry the ESD current without entering the avalanche breakdown region of the ESD clamp **120**. In some embodiments, ESD clamp **120** is implemented without having avalanching junctions inside ESD clamp **120**, and is also known as a “non-snapback protection scheme.”

[0044] Other types of clamp circuits, configurations and arrangements of ESD clamp **120** are within the scope of the present disclosure.

[0045] Other configurations or quantities of circuits in integrated circuit **100** are within the scope of the present disclosure.

[0046] In some embodiments, during an ESD event at the first node or the second node, the ESD clamp **120** is configured to discharge an ESD current of the ESD event in the forward ESD direction from the second node to the first node, in accordance with some embodiments. In comparison with other approaches in the forward ESD direction, integrated circuit **100** has better ESD discharging capability and performance while occupying less area.

[0047] FIG. 2A is a circuit diagram of an integrated circuit **200A**, in accordance with some embodiments.

[0048] Integrated circuit **200A** is an embodiment of ESD clamp **120**, and similar detailed description is therefore omitted.

[0049] Components that are the same or similar to those in one or more of FIGS. **1**, **2A-2E**, **3A-3E**, **4A-4E**, **5A-5E**, **6A-6B**, **7A-7B** and **8A-8B** (shown below) are given the same reference numbers, and detailed description thereof is thus omitted.

[0050] Node **Nd1** in FIGS. **2A-2E**, **3A-3E**, **4A-4E**, **5A-5E**, **6A-6B**, **7A-7B** and **8A-8B** corresponds to voltage supply node **104** of FIG. **1**. Node **Nd2** of FIGS. **2A-2E**, **3A-3E**, **4A-4E**, **5A-5E**, **6A-6B**, **7A-7B** and **8A-8B** corresponds to reference voltage supply node **106** of FIG. **1**.

[0051] Integrated circuit **200A** includes a control circuit **201**, an ESD detection circuit **202**, an ESD assist circuit **204a** and a discharging circuit **210**.

[0052] Control circuit **201** is coupled to ESD detection circuit **202**, ESD assist circuit **204a**, and discharging circuit **210** by nodes **Nd1** and **Nd2**. Control circuit **201** is further coupled between node **Nd1** and node **Nd2**. Control circuit **201** is configured to control at least ESD detection circuit **202**.

[0053] ESD detection circuit **202** is coupled to control circuit **201**, ESD assist circuit **204a**, discharging circuit **210** and a node **Nd3**. ESD detection circuit **202** is further coupled between node **Nd1** and node **Nd2**. ESD detection circuit **202** is configured to detect an ESD event at node **Nd2** (e.g., ESD current **I1** in the forward ESD direction), and to charge node **Nd3** in response to the ESD event, thereby turning on discharging circuit **210**. In some embodiments, in response to being turned on, discharging circuit **210** couples node **Nd1** and **Nd2** thereby providing an ESD discharge path between node **Nd1** and **Nd2**. In some embodiments, ESD detection circuit **202** is configured to detect an ESD event at node **Nd1** (e.g., ESD current **I2** in the reverse ESD direction).

[0054] ESD assist circuit **204a** is coupled to node **Nd1**, node **Nd3**, ESD detection circuit **202** and discharging circuit **210**. In response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), ESD assist circuit **204a** is configured to turn-on and clamp a voltage at node **Nd3** to be equal to a voltage of node **Nd1** (e.g., **VDD**) thereby assisting the detection circuit **204** in causing the discharging circuit **210** to turn-on. In some embodiments, in response to being turned on, discharging circuit **210** couples node **Nd2** and **Nd1** thereby providing a forward ESD discharge path between node **Nd2** and **Nd1**.

[0055] Discharging circuit **210** is coupled between node **Nd1** and node **Nd2**. Discharging circuit **210** is further coupled to node **Nd3**, ESD detection circuit **202** and ESD assist circuit **204a**. Discharging circuit **210** is configured to couple node **Nd1** and **Nd2** during an ESD event at node **Nd1** or node **Nd2**, thereby providing an ESD discharge path between node **Nd1** and **Nd2**.

[0056] ESD detection circuit **202** includes a resistor **R1**, a capacitor **C1**, an N-type Metal Oxide Semiconductor (NMOS) transistor **N1** and a P-type Metal Oxide Semiconductor (PMOS) transistor **P1**.

[0057] Discharging circuit **210** includes a PMOS transistor **P2**. PMOS transistor **P2** includes a gate, a drain and a source (not labelled).

[0058] ESD assist circuit **204a** includes a PMOS transistor **P3**. In some embodiments, PMOS transistor **P3** is a grounded gate PMOS (ggPMOS) transistor. PMOS transistor **P3** includes a gate, a drain and a source (not labelled).

[0059] Each of a first end of resistor **R1**, node **Nd2**, a source of NMOS transistor **N1**, a body of NMOS transistor **N1** and a drain of PMOS transistor **P2** are coupled together.

[0060] Each of a second end of resistor **R1**, node **Nd4**, a first end of capacitor **C1**, a gate of PMOS transistor **P1**, and a gate of NMOS transistor **N2** are coupled together.

[0061] Each of a second end of capacitor **C1**, node **Nd1**, a source of PMOS transistor **P1**, a body of PMOS transistor **P1**, a source of PMOS transistor **P2**, a body of PMOS transistor **P2**, a source of PMOS transistor **P3**, a gate of PMOS transistor **P3**, and a body of PMOS transistor **P3** are coupled together.

[0062] Each of a node **Nd3**, a drain of NMOS transistor **N1**, a drain of PMOS transistor **P1**, a source of PMOS transistor **P3** and a gate of PMOS transistor **P2** are coupled together.

[0063] In some embodiments, capacitor C1 is a transistor-coupled capacitor. For example in some embodiments, capacitor C1 is a transistor having a drain and source coupled together thereby forming a transistor-coupled capacitor.

[0064] Resistor R1 and capacitor C1 are configured as an RC network. Depending upon a location of an output of the RC network, the RC network is configured as either a low pass filter or a high pass filter.

[0065] NMOS transistor N1 and PMOS transistor P1 are configured as an inverter (not labelled). Thus, a slowly rising voltage at node Nd4 will be inverted by NMOS transistor N1 and PMOS transistor P1 (e.g., an inverter) thereby causing node Nd3 to rapidly rise. Furthermore, a rapidly rising voltage at node Nd4 will be inverted by NMOS transistor N1 and PMOS transistor P1 (e.g., an inverter) thereby causing node Nd3 to rise slowly. In some embodiments, NMOS transistor N1 and PMOS transistor P1 are configured to generate an inverted input signal (not shown) in response to an input signal (not shown).

[0066] When an ESD event at node Nd2 occurs (e.g., ESD current I1 in the forward ESD direction), the ESD current or voltage at node Nd2 rises rapidly causing the voltage of node Nd4 (e.g., across resistor R1) to rise rapidly since the voltage at node Nd4 corresponds to an output voltage of a high pass filter (e.g., a voltage across resistor R1 with respect to node Nd2). In other words, resistor R1 is configured as a high pass filter, and the rapidly changing voltage or current from the ESD event is not filtered or is passed by resistor R1. In response to the rapidly rising voltage at node Nd4, PMOS transistor P1 and NMOS transistor N1 are configured as an inverter thereby inverting the voltage of node Nd4 to the voltage of node Nd3, and the voltage of Nd3 is rapidly falling. Stated differently, the voltage of node Nd3 is inverted from the voltage of node Nd4. In some embodiments, in response to the voltage at node Nd4, PMOS transistor P3 turns on, and is configured to clamp a voltage at node Nd3 to be equal to a voltage of node Nd1 (e.g., VDD) thereby assisting the ESD detection circuit 202 in causing the PMOS transistor P2 to turn-on. In response to being turned on, PMOS transistor P2 couples node Nd2 to node Nd1 and NMOS transistor N2 discharges the ESD current I1 in the forward ESD direction from node Nd2 to Nd1.

[0067] In some embodiments, by including ESD assist circuit 204a in integrated circuit 200A, the PMOS transistor P3 of ESD assist circuit 204a is configured to control the gate potential of PMOS transistor P2 by clamping the voltage of node Nd3 to be equal to the voltage of node Nd1 (e.g., VDD) during an ESD event at node Nd2 thereby reducing capacitive coupling effects between node Nd2 and Nd3 and effectively lowering the turn-on resistance of PMOS transistor P2, and thus causing the PMOS transistor P2 to turn-on faster and stronger than other approaches resulting in improved ESD performance and improved ESD robustness compared to other approaches.

[0068] In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), a body diode of the PMOS transistor P3 is configured to provide ESD protection of integrated circuit 200A by clamping the voltage of node Nd3. In some embodiments, a body diode of the PMOS transistor P2 is configured to provide ESD protection of integrated circuit 200A by passing the ESD current I1 in the forward ESD direction from node Nd2 to node Nd1.

[0069] In some embodiments, ESD assist circuit 204a has a minimal effect on an ESD event at node Nd1. For example, in some embodiments, when an ESD event at node Nd1 occurs, PMOS transistor P3 is turned off.

[0070] Other types of circuits, configurations and arrangements of at least ESD detection circuit 202, ESD assist circuit 204a or discharging circuit 210 are within the scope of the present disclosure.

[0071] Other configurations or quantities of circuits in integrated circuit 200A are within the scope of the present disclosure.

[0072] FIG. 2B is a circuit diagram of an integrated circuit 200B, in accordance with some embodiments.

[0073] Integrated circuit **200B** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted.

[0074] Integrated circuit **200B** is a variation of integrated circuit **200A** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with integrated circuit **200A**, ESD assist circuit **204b** of integrated circuit **200B** replaces ESD assist circuit **204a** of integrated circuit **200A**, and similar detailed description is therefore omitted.

[0075] Integrated circuit **200B** includes control circuit **201**, ESD detection circuit **202**, ESD assist circuit **204b** and discharging circuit **210**.

[0076] ESD assist circuit **204b** is a variation of ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with ESD assist circuit **204a**, a diode D1 of ESD assist circuit **204b** replaces PMOS transistor P3 of ESD assist circuit **204a**, and similar detailed description is therefore omitted.

[0077] ESD assist circuit **204b** includes diode D1. Diode D1 includes an anode and a cathode (not labelled).

[0078] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the anode of diode D1, node Nd3, a drain of NMOS transistor N1, a drain of PMOS transistor P1 and a gate of PMOS transistor P2 are coupled together.

[0079] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the cathode of diode D1, the second end of capacitor C1, node Nd1, the source of PMOS transistor P1, the body of PMOS transistor P1, the source of PMOS transistor P2 and the body of PMOS transistor P2 are coupled together.

[0080] The description for when an ESD event occurs at node Nd2 (e.g., ESD current I1 in the forward ESD direction) with ESD assist circuit **204b** for FIG. 2B is similar to the description of when an ESD event occurs at node Nd2 for ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), diode D1 is configured to be forward-biased and thereby clamps the voltage at node Nd3 to be equal to a voltage of node Nd1 (e.g., VDD) thereby assisting the ESD detection circuit **202** in causing the PMOS transistor P2 to turn-on. In response to being turned on, PMOS transistor P2 couples node Nd2 to node Nd1 and PMOS transistor P2 discharges the ESD current I1 in the forward ESD direction from node Nd2 to Nd1.

[0081] In some embodiments, by including ESD assist circuit **204b** in integrated circuit **200B**, integrated circuit **200B** achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0082] Other types of circuits, configurations and arrangements of at least ESD detection circuit **202**, ESD assist circuit **204b** or discharging circuit **210** are within the scope of the present disclosure.

[0083] Other configurations or quantities of circuits in integrated circuit **200B** are within the scope of the present disclosure.

[0084] FIG. 2C is a circuit diagram of an integrated circuit **200C**, in accordance with some embodiments.

[0085] Integrated circuit **200C** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted.

[0086] Integrated circuit **200C** is a variation of integrated circuit **200A** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with integrated circuit **200A**, ESD assist circuit **204c** of integrated circuit **200C** replaces ESD assist circuit **204a** of integrated circuit **200A**, and similar detailed description is therefore omitted.

[0087] Integrated circuit **200C** includes control circuit **201**, ESD detection circuit **202**, ESD assist circuit **204c** and discharging circuit **210**.

[0088] ESD assist circuit **204c** is a variation of ESD assist circuit **204a** of FIG. 2A, and similar

detailed description is therefore omitted. In comparison with ESD assist circuit **204a**, a transistor **B1** of ESD assist circuit **204c** replaces PMOS transistor **P3** of ESD assist circuit **204a**, and similar detailed description is therefore omitted. Transistor **B1** is a PNP bipolar junction transistor (BJT). [0089] ESD assist circuit **204c** includes transistor **B1**. Transistor **B1** includes a base, a collector and an emitter (not labelled).

[0090] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the collector of transistor **B1**, node **Nd3**, a drain of NMOS transistor **N1**, a drain of PMOS transistor **P1** and a gate of PMOS transistor **P2** are coupled together.

[0091] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the emitter of transistor **B1**, the base of transistor **B1**, the second end of capacitor **C1**, node **Nd1**, the source of PMOS transistor **P1**, the body of PMOS transistor **P1**, the source of PMOS transistor **P2** and the body of PMOS transistor **P2** are coupled together.

[0092] The description for when an ESD event occurs at node **Nd2** (e.g., ESD current **I1** in the forward ESD direction) with ESD assist circuit **204c** for FIG. 2C is similar to the description of when an ESD event occurs at node **Nd2** for ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), transistor **B1** turns on, and is configured to clamp a voltage at node **Nd3** to be equal to a voltage of node **Nd1** (e.g., **VDD**) thereby assisting the ESD detection circuit **202** in causing the PMOS transistor **P2** to turn-on. In response to being turned on, PMOS transistor **P2** couples node **Nd2** to node **Nd1**, and PMOS transistor **P2** discharges the ESD current **I1** in the forward ESD direction from node **Nd2** to **Nd1**.

[0093] In some embodiments, in response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), a body diode of transistor **B1** is configured to provide ESD protection of integrated circuit **200C** by clamping the voltage of node **Nd3**. In some embodiments, a body diode of PMOS transistor **P2** is configured to provide ESD protection of integrated circuit **200C** by passing the ESD current **I1** in the forward ESD direction from node **Nd2** to node **Nd1**.

[0094] In some embodiments, by including ESD assist circuit **204c** in integrated circuit **200C**, integrated circuit **200C** achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0095] Other types of circuits, configurations and arrangements of at least ESD detection circuit **202**, ESD assist circuit **204c** or discharging circuit **210** are within the scope of the present disclosure.

[0096] Other configurations or quantities of circuits in integrated circuit **200C** are within the scope of the present disclosure.

[0097] FIG. 2D is a circuit diagram of an integrated circuit **200D**, in accordance with some embodiments.

[0098] Integrated circuit **200D** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted.

[0099] Integrated circuit **200D** is a variation of integrated circuit **200A** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with integrated circuit **200A**, ESD assist circuit **204d** of integrated circuit **200D** replaces ESD assist circuit **204a** of integrated circuit **200A**, and similar detailed description is therefore omitted.

[0100] Integrated circuit **200D** includes control circuit **201**, ESD detection circuit **202**, ESD assist circuit **204d** and discharging circuit **210**.

[0101] ESD assist circuit **204d** is a variation of ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with ESD assist circuit **204a**, an NMOS transistor **N2** of ESD assist circuit **204d** replaces PMOS transistor **P3** of ESD assist circuit **204a**, and similar detailed description is therefore omitted.

[0102] ESD assist circuit **204d** includes NMOS transistor **N2**. NMOS transistor **N2** includes a gate, a drain, a source and a body (not labelled).

[0103] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the gate of NMOS transistor **N2**, the source of NMOS transistor **N2**, the body of NMOS transistor **N2**, node **Nd3**, a drain of NMOS transistor **N1**, a drain of PMOS transistor **P1** and a gate of PMOS transistor **P2** are coupled together.

[0104] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the drain of NMOS transistor **N2**, the second end of capacitor **C1**, node **Nd1**, the source of PMOS transistor **P1**, the body of PMOS transistor **P1**, the source of PMOS transistor **P2** and the body of PMOS transistor **P2** are coupled together.

[0105] The description for when an ESD event occurs at node **Nd2** (e.g., ESD current **I1** in the forward ESD direction) with ESD assist circuit **204d** for FIG. 2D is similar to the description of when an ESD event occurs at node **Nd2** for ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), NMOS transistor **N2** turns on, and is configured to clamp a voltage at node **Nd3** to be equal to a voltage of node **Nd1** (e.g., **VDD**) thereby assisting the ESD detection circuit **202** in causing the PMOS transistor **P2** to turn-on. In response to being turned on, PMOS transistor **P2** couples node **Nd2** to node **Nd1**, and PMOS transistor **P2** discharges the ESD current **I1** in the forward ESD direction from node **Nd2** to **Nd1**.

[0106] In some embodiments, in response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), a body diode of NMOS transistor **N2** is configured to provide ESD protection of integrated circuit **200D** by clamping the voltage of node **Nd3**. In some embodiments, a body diode of PMOS transistor **P2** is configured to provide ESD protection of integrated circuit **200D** by passing the ESD current **I1** in the forward ESD direction from node **Nd2** to node **Nd1**.

[0107] In some embodiments, by including ESD assist circuit **204d** in integrated circuit **200D**, integrated circuit **200D** achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0108] Other types of circuits, configurations and arrangements of at least ESD detection circuit **202**, ESD assist circuit **204d** or discharging circuit **210** are within the scope of the present disclosure.

[0109] Other configurations or quantities of circuits in integrated circuit **200D** are within the scope of the present disclosure.

[0110] FIG. 2E is a circuit diagram of an integrated circuit **200E**, in accordance with some embodiments.

[0111] Integrated circuit **200E** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted.

[0112] Integrated circuit **200E** is a variation of integrated circuit **200A** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with integrated circuit **200A**, ESD assist circuit **204e** of integrated circuit **200E** replaces ESD assist circuit **204a** of integrated circuit **200A**, and similar detailed description is therefore omitted.

[0113] Integrated circuit **200E** includes control circuit **201**, ESD detection circuit **202**, ESD assist circuit **204e** and discharging circuit **210**.

[0114] ESD assist circuit **204e** is a variation of ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with ESD assist circuit **204a**, a transistor **B2** of ESD assist circuit **204e** replaces PMOS transistor **P3** of ESD assist circuit **204a**, and similar detailed description is therefore omitted. Transistor **B2** is a NPN BJT.

[0115] ESD assist circuit **204e** includes transistor **B2**. Transistor **B2** includes a base, a collector and an emitter (not labelled).

[0116] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the emitter of transistor **B2**, the base of transistor **B2**, node **Nd3**, a drain of NMOS transistor **N1**, a drain of PMOS transistor **P1** and a gate of PMOS transistor **P2** are coupled together.

[0117] In comparison with ESD assist circuit **204a** of FIG. 2A, each of the collector of transistor **B2**, the second end of capacitor **C1**, node **Nd1**, the source of PMOS transistor **P1**, the body of PMOS transistor **P1**, the source of PMOS transistor **P2** and the body of PMOS transistor **P2** are coupled together.

[0118] The description for when an ESD event occurs at node **Nd2** (e.g., ESD current **I1** in the forward ESD direction) with ESD assist circuit **204e** for FIG. 2E is similar to the description of when an ESD event occurs at node **Nd2** for ESD assist circuits **204a** and **204c** of corresponding FIGS. 2A and 2C, and similar detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), transistor **B2** turns on, and is configured to clamp a voltage at node **Nd3** to be equal to a voltage of node **Nd1** (e.g., **VDD**) thereby assisting the ESD detection circuit **202** in causing the PMOS transistor **P2** to turn-on. In response to being turned on, PMOS transistor **P2** couples node **Nd2** to node **Nd1**, and PMOS transistor **P2** discharges the ESD current **I1** in the forward ESD direction from node **Nd2** to **Nd1**.

[0119] In some embodiments, in response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), a body diode of transistor **B2** is configured to provide ESD protection of integrated circuit **200E** by clamping the voltage of node **Nd3**. In some embodiments, a body diode of PMOS transistor **P2** is configured to provide ESD protection of integrated circuit **200E** by passing the ESD current **I1** in the forward ESD direction from node **Nd2** to node **Nd1**.

[0120] In some embodiments, by including ESD assist circuit **204e** in integrated circuit **200E**, integrated circuit **200E** achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0121] Other types of circuits, configurations and arrangements of at least ESD detection circuit **202**, ESD assist circuit **204e** or discharging circuit **210** are within the scope of the present disclosure.

[0122] Other configurations or quantities of circuits in integrated circuit **200E** are within the scope of the present disclosure.

[0123] FIG. 3A is a cross-sectional view of an integrated circuit **300A**, in accordance with some embodiments.

[0124] Integrated circuit **300A** is an embodiment of PMOS transistor **P3** of ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted.

[0125] Integrated circuit **300A** includes a substrate **302b**.

[0126] Substrate **302b** is an n-type substrate. In some embodiments, substrate **302b** is a p-type substrate. In some embodiments, substrate **302b** includes an elemental semiconductor including silicon or germanium in crystal, polycrystalline, or an amorphous structure; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and indium antimonide; an alloy semiconductor including SiGe, GaAsP, AlInAs, AlGaAs, GaInAs, GaInP, and GaInAsP; any other suitable material; or combinations thereof. In some embodiments, the alloy semiconductor substrate has a gradient SiGe feature in which the Si and Ge composition change from one ratio at one location to another ratio at another location of the gradient SiGe feature. In some embodiments, the alloy SiGe is formed over a silicon substrate. In some embodiments, substrate **302b** is a strained SiGe substrate. In some embodiments, the semiconductor substrate has a semiconductor on insulator structure, such as a silicon on insulator (SOI) structure. In some embodiments, the semiconductor substrate includes a doped epi layer or a buried layer. In some embodiments, the compound semiconductor substrate has a multilayer structure, or the substrate includes a multilayer compound semiconductor structure.

[0127] In some embodiments, substrate **302b** is replaced with an N-well that is formed inside a p-type substrate. In some embodiments, the N-well has N-type dopants, and the p-type substrate has p-type dopants.

[0128] Integrated circuit **300A** further includes a gate structure **330** that is over substrate **302b**. The

gate structure **330** includes a gate dielectric **322** and a gate electrode **330a**. Integrated circuit **300A** further includes sidewalls **320a** and **320b** on opposite sides of gate structure **330**.

[0129] Integrated circuit **300A** further includes a drain region **312a** and a source region **314a**. The source region **314a** is a P+ active region having P-type dopants implanted in the well (not shown) or substrate **302b**. The drain region **312a** is a P+ active region having P-type dopants implanted in well (not shown) or substrate **302b**. In some embodiments, at least source region **314a** or drain region **312a** extends above substrate **302b**.

[0130] In some embodiments, the drain region **312a** and source region **314a** of transistor **360a** of FIG. 3A or active regions of other devices of FIGS. 2B-2E and 4A-4E are referred to as an oxide definition (OD) region which defines active regions of integrated circuits **300A-300E** and **400A-400E** (described below).

[0131] In some embodiments, integrated circuit **300A** further includes lightly doped drain (LDD) regions **318** adjacent to source region **314a** and drain region **312a**, and below sidewalls **320**. In some embodiments, a first element and a second element are adjacent to each other includes scenarios where the first element and the second element are directly next to each other. In some embodiments, a first element and a second element are adjacent to each other includes scenarios where intermediary elements are positioned between the first element and the second element. In some embodiments, LDD region **360** helps transistor **360a** maintain a low leakage current.

[0132] In some embodiments, drain region **312a**, source region **314a**, LDD regions **318**, sidewalls **320** and gate structure **330** together form a transistor **360a**. Transistor **360a** is a PMOS transistor.

[0133] Drain region **312a** corresponds to the drain of PMOS transistor P3 of FIG. 2A, source region **314a** corresponds to the source of PMOS transistor P3 of FIG. 2A, gate structure **330** corresponds to the gate of PMOS transistor P3 of FIG. 2A, and tap regions **316a** and **317a** correspond to the body of PMOS transistor P3 of FIG. 2A, and similar detailed description is therefore omitted.

[0134] The drain region **312a** is coupled to node Nd3 (shown in FIG. 2A). The gate structure **330**, the source region **314a** and the tap regions **316a** and **316b** are coupled to the supply voltage (e.g., voltage VDD) by node Nd1. In some embodiments, each of the gate structure **330**, source region **314a** and tap regions **316a** and **316b** are coupled together at node Nd1, and are further coupled to the supply voltage (e.g., voltage VDD).

[0135] In some embodiments, drain region **312a** and source region **314a** comprise fins in accordance with fin field-effect transistor (FinFET) complementary metal-oxide-semiconductor (CMOS) technologies. In some embodiments, drain region **312a** and source region **314a** comprise nanosheets of nanosheet transistors. In some embodiments, drain region **312a** and source region **314a** comprise nanowire of nanowire transistors. In some embodiments, drain region **312a** and source region **314a** are free of fins in accordance with planar CMOS technologies. Other types of transistors are within the scope of the present disclosure.

[0136] In some embodiments, drain region **312a** is an extended drain region and has a greater size than source region **314a**. In at least one embodiment, a silicide layer (not shown) covers a portion, but not the entirety, of drain region **312a**. Such a partially silicided configuration of drain region **312a** improves self-protection of transistor **360a** from ESD events. In at least one embodiment, drain region **312a** is fully silicided.

[0137] Gate structure **330** is arranged between drain region **312a** and source region **314a**. In some embodiments, gate electrode **330a** comprises a conductive material such as metal or polysilicon (also referred to herein as "POLY"). In some embodiments, gate structure **330** is coupled to a supply voltage terminal (e.g., voltage VDD), or a reference supply voltage terminal (e.g., voltage VSS).

[0138] Integrated circuit **300A** further includes tap regions **316a** and **317a** and shallow trench isolation (STI) regions **308**, **309**, **310** and **311**.

[0139] STI region **308** is configured to isolate at least drain region **312a** and tap region **317a** from

each other. In some embodiments, STI region **308** is configured to isolate drain region **312a** or tap region **317a** from other portions of integrated circuit **300A** (not shown).

[0140] STI region **309** is configured to isolate tap region **317a** from other portions of integrated circuit **300A**.

[0141] STI region **310** is configured to isolate at least source region **314a** and tap region **316a** from each other. In some embodiments, STI region **310** is configured to isolate source region **314a** or tap region **316a** from other portions of integrated circuit **300A** (not shown).

[0142] STI region **311** is configured to isolate tap region **316a** from other portions of integrated circuit **300A**.

[0143] In some embodiments, tap regions **316a** and **317a** correspond to a well tap. In some embodiments, a well tap is electrically conductive materials that couple source/drain regions of integrated circuit **300A** to a voltage supply node (e.g., supply voltage VDD).

[0144] Tap regions **316a** and **317a** are heavily doped n-regions in a well (not shown) or substrate **302b**. In some embodiments, tap regions **316a** and **317a** are heavily doped n-regions in a n-type well (not shown) on a p-type substrate. In some embodiments, the heavily doped n-region is coupled through the well tap to the voltage supply node (e.g., supply voltage VDD) thereby setting the potential of the n-type well or substrate **302b** to prevent leakage from adjacent source/drain regions into the n-well/p-substrate.

[0145] Integrated circuit **300A** further includes a body diode **340a**. Body diode **340a** is a parasitic element formed by the PN junction between the drain region **312a** and at least tap region **316a** or **317a**. In some embodiments, the body diode **340a** is formed by the PN junction between the drain region **312a** and the source region **314a**. The body diode **340a** is also referred to as a parasitic diode or an internal diode of integrated circuit **300A**. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode **340a** of PMOS transistor P3 is configured to provide ESD protection of integrated circuit **300A** by clamping the voltage of node Nd3, and integrated circuit **300A** thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0146] FIG. 3B is a cross-sectional view of an integrated circuit **300B**, in accordance with some embodiments.

[0147] Integrated circuit **300B** is an embodiment of diode D1 of ESD assist circuit **204b** of FIG. 2B, and similar detailed description is therefore omitted. Integrated circuit **300B** is a variation of integrated circuit **300A** of FIG. 3A, and similar detailed description is therefore omitted.

[0148] Integrated circuit **300B** includes a substrate **302b**, a P-type region **352a**, an N-type region **356a**, and STI regions **308**, **310** and **311**.

[0149] P-type region **352a** is similar to the drain region **312a** in FIG. 3A, and similar detailed description is therefore omitted.

[0150] Substrate **302b** is an n-type substrate having N-type dopants. The P-type region **352a** is a P+ active region having P-type dopants implanted in the well (not shown) or substrate **302b**. The N-type region **356a** is an N+ active region having N-type dopants implanted in the well (not shown) or substrate **302b**. STI region **310** is between P-type region **352a** and N-type region **356a**.

[0151] STI region **308** is configured to isolate at least P-type region **352a** from other portions of integrated circuit **300B** (not shown). STI region **311** is configured to isolate at least N-type region **356a** from other portions of integrated circuit **300B** (not shown).

[0152] STI region **310** is configured to isolate at least P-type region **352a** and N-type region **356a** from each other. In some embodiments, STI region **310** is configured to isolate P-type region **352a** or N-type region **356a** from other portions of integrated circuit **300B** (not shown).

[0153] P-type region **352a** corresponds to the anode of diode D1 of FIG. 2B, and N-type region **356a** corresponds to the cathode of diode D1 of FIG. 2B, and similar detailed description is therefore omitted.

[0154] The P-type region **352a** is coupled to node Nd3 (shown in FIG. 2A). The N-type region

356a is coupled to the supply voltage (e.g., voltage VDD) by node Nd1.

[0155] Integrated circuit **300B** further includes a body diode **340b**. Body diode **340b** is a parasitic element formed by the PN junction between the P-type region **352a** and N-type region **356a**. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode **340b** is configured to provide ESD protection of integrated circuit **300B** by clamping the voltage of node Nd3, and integrated circuit **300B** thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0156] FIG. 3C is a cross-sectional view of an integrated circuit **300C**, in accordance with some embodiments.

[0157] Integrated circuit **300C** is an embodiment of transistor B1 of ESD assist circuit **204c** of FIG. 2C, and similar detailed description is therefore omitted. Integrated circuit **300C** is a variation of integrated circuit **300A** of FIG. 3A, and similar detailed description is therefore omitted.

[0158] Integrated circuit **300C** includes a substrate **302a**, a P-type region **362a**, and an N-type region **366a**. Substrate **302a** is similar to substrate **302b**, P-type region **362a** is similar to P-type region **352a** in FIG. 3B, N-type region **366a** is similar to N-type region **356a** in FIG. 3B, and similar detailed description is therefore omitted.

[0159] In FIG. 3C, substrate **302a** is a p-type substrate having P-type dopants. The N-type region **366a** is an N+ active region having N-type dopants implanted in substrate **302a**. The P-type region **362a** is a P+ active region having P-type dopants implanted in the N-type region **366a**.

[0160] P-type region **362a** corresponds to the emitter of transistor B1 of FIG. 2C, N-type region **366a** corresponds to the base of transistor B1 of FIG. 2C, substrate **302a** corresponds to the collector of transistor B1 of FIG. 2C, and similar detailed description is therefore omitted.

[0161] Each of the P-type region **362a** and the N-type region **366a** are coupled together, and further coupled to the supply voltage (e.g., voltage VDD) by node Nd1. The substrate **302a** is coupled to node Nd3.

[0162] Integrated circuit **300C** further includes a body diode **340c**. Body diode **340c** is a parasitic element formed by the PN junction between the substrate **302a** and the N-type region **366a**. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode **340c** is configured to provide ESD protection of integrated circuit **300C** by clamping the voltage of node Nd3, and integrated circuit **300C** thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0163] FIG. 3D is a cross-sectional view of an integrated circuit **300D**, in accordance with some embodiments.

[0164] Integrated circuit **300D** is an embodiment of NMOS transistor N2 of ESD assist circuit **204d** of FIG. 2D, and similar detailed description is therefore omitted. Integrated circuit **300D** is a variation of integrated circuit **300A** of FIG. 3A, and similar detailed description is therefore omitted.

[0165] In comparison with integrated circuit **300A** of FIG. 3A, a drain region **312b** replaces drain region **312a** of FIG. 3A, a source region **314b** replaces source region **314a** of FIG. 3A, tap regions **316b** and **317b** replace corresponding tap regions **316a** and **317a** of FIG. 3A, transistor **360b** replaces transistor **360a**, and similar detailed description is therefore omitted.

[0166] Integrated circuit **300D** includes substrate **302a**, gate structure **330**, drain region **312b**, source region **314b**, LDD regions **318**, sidewalls **320**, tap regions **316b** and **317b**, and STI regions **308**, **309**, **310** and **311**, and body diode **340d**.

[0167] In FIG. 3D, substrate **302a** is a p-type substrate having P-type dopants.

[0168] In some embodiments, drain region **312b**, source region **314b**, LDD regions **318**, sidewalls **320** and gate structure **330** together form a transistor **360b**. Transistor **360b** is an NMOS transistor.

[0169] Drain region **312b** corresponds to the drain of NMOS transistor N2 of FIG. 2D, source region **314b** corresponds to the source of NMOS transistor N2 of FIG. 2D, gate structure **330** corresponds to the gate of NMOS transistor N2 of FIG. 2D, and tap regions **316b** and **317b**

correspond to the body of NMOS transistor N2 of FIG. 2D, and similar detailed description is therefore omitted.

[0170] The drain region **312b** is coupled to the supply voltage (e.g., voltage VDD) by node Nd1. The gate structure **330**, the source region **314b** and the tap region **316b** are coupled to node Nd3.

[0171] STI region **308** is configured to isolate at least drain region **312b** and tap region **317b** from each other. In some embodiments, STI region **308** is configured to isolate drain region **312b** or tap region **317b** from other portions of integrated circuit **300D** (not shown).

[0172] STI region **309** is configured to isolate tap region **317b** from other portions of integrated circuit **300D**.

[0173] STI region **310** is configured to isolate at least source region **314b** and tap region **316b** from each other. In some embodiments, STI region **310** is configured to isolate source region **314b** or tap region **316b** from other portions of integrated circuit **300D** (not shown).

[0174] STI region **311** is configured to isolate tap region **316b** from other portions of integrated circuit **300D**.

[0175] In some embodiments, tap regions **316b** and **317b** correspond to a substrate tap. In some embodiments, a substrate tap is an electrically conductive material that couples tap regions **316b** and **317b** to a reference voltage supply node (e.g., supply voltage VSS). For example, in some embodiments, tap regions **316b** and **317b** of substrate **302a** includes a heavily doped p-region which is formed in a p-type substrate. In some embodiments, the heavily doped p-region is coupled through the tap regions **316b** and **317b** to the reference voltage supply node (e.g., reference supply voltage VSS) thereby setting the potential of the substrate **302a** to prevent leakage from adjacent source/drain regions.

[0176] Integrated circuit **300D** further includes a body diode **340d**. Body diode **340d** is a parasitic element formed by the PN junction between at least tap region **316b** or **317b** and the drain region **312b**. In some embodiments, the body diode **340d** is formed by the PN junction between the source region **314b** and the drain region **312b**. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode **340d** of NMOS transistor N2 is configured to provide ESD protection of integrated circuit **300D** by clamping the voltage of node Nd3, and integrated circuit **300D** thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0177] FIG. 3E is a cross-sectional view of an integrated circuit **300E**, in accordance with some embodiments.

[0178] Integrated circuit **300E** is an embodiment of transistor B2 of ESD assist circuit **204e** of FIG. 2E, and similar detailed description is therefore omitted. Integrated circuit **300E** is a variation of integrated circuit **300C** of FIG. 3C, and similar detailed description is therefore omitted.

[0179] Integrated circuit **300E** includes substrate **302b**, an N-type region **362b**, and a P-type region **366b**. N-type region **362b** is similar to N-type region **366a** in FIG. 3B, P-type region **366b** is similar to P-type region **362a** in FIG. 3B, and similar detailed description is therefore omitted.

[0180] Substrate **302b** is an n-type substrate having N-type dopants. The P-type region **366b** is a P+ active region having P-type dopants implanted in substrate **302b**. The N-type region **362b** is an N+ active region having N-type dopants implanted in the P-type region **366b**.

[0181] N-type region **362b** corresponds to the emitter of transistor B2 of FIG. 2E, P-type region **366b** corresponds to the base of transistor B2 of FIG. 2E, substrate **302b** corresponds to the collector of transistor B2 of FIG. 2E, and similar detailed description is therefore omitted.

[0182] Each of the N-type region **362b** and the P-type region **366b** are coupled together, and further coupled to node Nd3. The substrate **302b** is coupled to the supply voltage (e.g., voltage VDD) by node Nd1.

[0183] Integrated circuit **300E** further includes a body diode **340e**. Body diode **340e** is a parasitic element formed by the PN junction between the P-type region **366b** and the substrate **302b**. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode

340e is configured to provide ESD protection of integrated circuit **300E** by clamping the voltage of node **Nd3**, and integrated circuit **300E** thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 2A.

[0184] FIG. 4A is a circuit diagram of an integrated circuit **400A**, in accordance with some embodiments.

[0185] Integrated circuit **400A** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted.

[0186] Integrated circuit **400A** is a variation of integrated circuit **200A** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with integrated circuit **200A**, ESD detection circuit **402** of integrated circuit **400A** replaces ESD detection circuit **202** of integrated circuit **200A**, ESD assist circuit **404a** of integrated circuit **400A** replaces ESD assist circuit **204a** of integrated circuit **200A**, discharging circuit **410** of integrated circuit **400A** replaces discharging circuit **210** of integrated circuit **200A**, and similar detailed description is therefore omitted.

[0187] Integrated circuit **400A** includes control circuit **201**, ESD detection circuit **402**, ESD assist circuit **404a** and discharging circuit **410**.

[0188] ESD detection circuit **402** is a variation of ESD detection circuit **202** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with ESD detection circuit **202**, a resistor **R2** of ESD detection circuit **402** replaces resistor **R1** of ESD detection circuit **202**, a capacitor **C2** of ESD detection circuit **402** replaces capacitor **C1** of ESD detection circuit **202**, and similar detailed description is therefore omitted.

[0189] ESD discharge circuit **410** is a variation of ESD discharge circuit **210** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with ESD discharge circuit **210**, an NMOS transistor **N3** of ESD discharge circuit **410** replaces PMOS transistor **P2** of ESD discharge circuit **210**, and similar detailed description is therefore omitted.

[0190] ESD assist circuit **404a** is a variation of ESD assist circuit **204a** of FIG. 2A, and similar detailed description is therefore omitted. In comparison with ESD assist circuit **204a**, an NMOS transistor **N4** of ESD assist circuit **404a** replaces PMOS transistor **P3** of ESD assist circuit **204a**, and similar detailed description is therefore omitted.

[0191] ESD assist circuit **404a** includes NMOS transistor **N4**. In some embodiments, NMOS transistor **N4** is a grounded gate NMOS (ggNMOS) transistor. NMOS transistor **N4** includes a gate, a drain, a source and a body (not labelled).

[0192] ESD discharge circuit **410** includes NMOS transistor **N3**. NMOS transistor **N3** includes a gate, a drain, a source and a body (not labelled).

[0193] Each of a first end of capacitor **C2**, node **Nd2**, a source of NMOS transistor **N1**, a body of NMOS transistor **N1**, source of NMOS transistor **N3**, a body of NMOS transistor **N3**, a gate of NMOS transistor **N4**, a source of NMOS transistor **N4**, and a body of NMOS transistor **N4** are coupled together.

[0194] Each of a first end of resistor **R2**, node **Nd4**, a second end of capacitor **C2**, the gate of PMOS transistor **P1**, and the gate of NMOS transistor **N2** are coupled together.

[0195] Each of a second end of resistor **R2**, node **Nd1**, the source of PMOS transistor **P1**, the body of PMOS transistor **P1**, and a drain of NMOS transistor **N3** are coupled together.

[0196] Each of a node **Nd3**, a drain of NMOS transistor **N1**, a drain of PMOS transistor **P1**, a drain of NMOS transistor **N4** and a gate of NMOS transistor **N3** are coupled together.

[0197] In some embodiments, capacitor **C2** is a transistor-coupled capacitor. For example in some embodiments, capacitor **C2** is a transistor having a drain and source coupled together thereby forming a transistor-coupled capacitor.

[0198] Resistor **R2** and capacitor **C2** are configured as an RC network. Dependent upon a location of an output of the RC network, the RC network is configured as either a low pass filter or a high pass filter.

[0199] When an ESD event at node **Nd2** occurs (e.g., ESD current **I1** in the forward ESD

direction), the ESD current or voltage at node Nd2 rises rapidly causing the voltage of node Nd4 (e.g., across capacitor C2) to rise slowly since the voltage at node Nd4 corresponds to an output voltage of a low pass filter (e.g., a voltage across capacitor C2 with respect to node Nd2). In other words, capacitor C2 is configured as a low pass filter, and the rapidly changing voltage or current from the ESD event is filtered by capacitor C2. In some embodiments, ESD detection circuit 402 has a minimal effect on an ESD event at node Nd2.

[0200] When an ESD event at node Nd2 occurs (e.g., ESD current I1 flows in the forward ESD direction), the ESD current or voltage at node Nd2 rises rapidly, and ESD assist circuit 404a detects the rapidly rising current or voltage at node Nd2 of the ESD event, causing NMOS transistor N4 of the ESD assist circuit 404a to turn on. In response to turning on, NMOS transistor N4 couples node Nd2 to node Nd3 and thereby charges node Nd3 and the gate of NMOS transistor N3 of the discharging circuit 410 in response to the rising ESD voltage or current. Stated differently, NMOS transistor N4 clamps the voltage of node Nd3 at the voltage of node Nd2 (VSS). In response to being charged by NMOS transistor N4 of ESD assist circuit 404a, NMOS transistor N3 of discharging circuit 410 is turned on and couples node Nd2 to node Nd1. By being turned on and coupling node Nd2 to node Nd1, NMOS transistor N3 discharges the ESD current I1 in the forward ESD direction from node Nd2 to Nd1.

[0201] In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), a body diode of the NMOS transistor N4 is configured to provide ESD protection of integrated circuit 400A by clamping the voltage of node Nd3. In some embodiments, a body diode of the NMOS transistor N3 is configured to provide ESD protection of integrated circuit 400A by passing the ESD current I1 in the forward ESD direction from node Nd2 to node Nd1.

[0202] In some embodiments, by including ESD assist circuit 404a in integrated circuit 400A, the NMOS transistor N4 of ESD assist circuit 404a is configured to control the gate potential of NMOS transistor N3 by clamping the voltage of node Nd3 to be equal to the voltage of node Nd2 (e.g., VSS) during an ESD event at node Nd2 thereby reducing capacitive coupling effects between node Nd1 and Nd3 and effectively lowering the turn-on resistance of NMOS transistor N3, and thus causing the NMOS transistor N3 to turn-on faster and stronger than other approaches resulting in improved ESD performance and improved ESD robustness compared to other approaches.

[0203] Other types of circuits, configurations and arrangements of at least ESD detection circuit 402, ESD assist circuit 404a or discharging circuit 410 are within the scope of the present disclosure.

[0204] Other configurations or quantities of circuits in integrated circuit 400A are within the scope of the present disclosure.

[0205] FIG. 4B is a circuit diagram of an integrated circuit 400B, in accordance with some embodiments.

[0206] Integrated circuit 400B is an embodiment of at least ESD clamp 120, and similar detailed description is therefore omitted.

[0207] Integrated circuit 400B is a variation of integrated circuit 400A of FIG. 4A, and similar detailed description is therefore omitted. In comparison with integrated circuit 400A, ESD assist circuit 404b of integrated circuit 400B replaces ESD assist circuit 404a of integrated circuit 400A, and similar detailed description is therefore omitted.

[0208] Integrated circuit 400B includes control circuit 201, ESD detection circuit 402, ESD assist circuit 404b and discharging circuit 410.

[0209] ESD assist circuit 404b is a variation of ESD assist circuit 404a of FIG. 4A, and similar detailed description is therefore omitted. In comparison with ESD assist circuit 404a, a diode D2 of ESD assist circuit 404b replaces NMOS transistor N4 of ESD assist circuit 404a, and similar detailed description is therefore omitted.

[0210] ESD assist circuit 404b includes diode D2. Diode D2 includes an anode and a cathode (not

labelled).

[0211] In comparison with ESD assist circuit **404a** of FIG. **4A**, each of the cathode of diode **D2**, node **Nd3**, the drain of NMOS transistor **N1**, the drain of PMOS transistor **P1**, and the gate of NMOS transistor **N3** are coupled together.

[0212] In comparison with ESD assist circuit **404a** of FIG. **4A**, each of the anode of diode **D2**, the first end of capacitor **C2**, node **Nd2**, the source of NMOS transistor **N1**, the body of NMOS transistor **N1**, the source of NMOS transistor **N3**, and the body of NMOS transistor **N3** are coupled together.

[0213] The description for when an ESD event occurs at node **Nd2** (e.g., ESD current **I1** in the forward ESD direction) with ESD assist circuit **404b** for FIG. **4B** is similar to the description of when an ESD event occurs at node **Nd2** for ESD assist circuit **404a** of FIG. **4A**, and similar detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node **Nd2** (e.g., an ESD current **I1** in the forward ESD direction), diode **D2** is configured to be forward-biased and thereby clamps the voltage at node **Nd3** to be equal to a voltage of node **Nd2** (e.g., **VSS**) thereby causing the NMOS transistor **N3** to turn-on. In response to being turned on, NMOS transistor **N3** couples node **Nd2** to node **Nd1** and NMOS transistor **N3** discharges the ESD current **I1** in the forward ESD direction from node **Nd2** to **Nd1**.

[0214] In some embodiments, by including ESD assist circuit **404b** in integrated circuit **400B**, integrated circuit **400B** achieves one or more of the benefits discussed above in at least FIG. **1** or FIG. **4A**.

[0215] Other types of circuits, configurations and arrangements of at least ESD detection circuit **402**, ESD assist circuit **404b** or discharging circuit **410** are within the scope of the present disclosure.

[0216] Other configurations or quantities of circuits in integrated circuit **400B** are within the scope of the present disclosure.

[0217] FIG. **4C** is a circuit diagram of an integrated circuit **400C**, in accordance with some embodiments.

[0218] Integrated circuit **400C** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted.

[0219] Integrated circuit **400C** is a variation of integrated circuit **400A** of FIG. **4A**, and similar circuit **404c** of integrated circuit **400C** replaces ESD assist circuit **404a** of integrated circuit **400A**, and similar detailed description is therefore omitted.

[0220] Integrated circuit **400C** includes control circuit **201**, ESD detection circuit **402**, ESD assist circuit **404c** and discharging circuit **410**.

[0221] ESD assist circuit **404c** is a variation of ESD assist circuit **404a** of FIG. **4A**, and similar detailed description is therefore omitted. In comparison with ESD assist circuit **404a**, a transistor **B3** of ESD assist circuit **404c** replaces NMOS transistor **N4** of ESD assist circuit **404a**, and similar detailed description is therefore omitted. Transistor **B3** is an NPN BJT.

[0222] ESD assist circuit **404c** includes transistor **B3**. Transistor **B3** includes a base, a collector and an emitter (not labelled).

[0223] In comparison with ESD assist circuit **404a** of FIG. **4A**, each of the collector of transistor **B3**, node **Nd3**, the drain of NMOS transistor **N1**, the drain of PMOS transistor **P1**, and the gate of NMOS transistor **N3** are coupled together.

[0224] In comparison with ESD assist circuit **404a** of FIG. **4A**, each of the emitter of transistor **B3**, the base of transistor **B3**, the first end of capacitor **C2**, node **Nd2**, the source of NMOS transistor **N1**, the body of NMOS transistor **N1**, the source of NMOS transistor **N3**, and the body of NMOS transistor **N3** are coupled together.

[0225] The description for when an ESD event occurs at node **Nd2** (e.g., ESD current **I1** in the forward ESD direction) with ESD assist circuit **404c** for FIG. **4C** is similar to the description of when an ESD event occurs at node **Nd2** for ESD assist circuit **404a** of FIG. **4A**, and similar

detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), transistor B3 turns on, and is configured to clamp the voltage at node Nd3 to be equal to a voltage of node Nd2 (e.g., VSS) thereby causing the NMOS transistor N3 to turn-on. In response to being turned on, NMOS transistor N3 couples node Nd2 to node Nd1 and NMOS transistor N3 discharges the ESD current I1 in the forward ESD direction from node Nd2 to Nd1.

[0226] In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), a body diode of transistor B3 is configured to provide ESD protection of integrated circuit 400C by clamping the voltage of node Nd3. In some embodiments, a body diode of NMOS transistor N3 is configured to provide ESD protection of integrated circuit 400B by passing the ESD current I1 in the forward ESD direction from node Nd2 to node Nd1.

[0227] In some embodiments, by including ESD assist circuit 404c in integrated circuit 400C, integrated circuit 400C achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 4A.

[0228] Other types of circuits, configurations and arrangements of at least ESD detection circuit 402, ESD assist circuit 404c or discharging circuit 410 are within the scope of the present disclosure.

[0229] Other configurations or quantities of circuits in integrated circuit 400C are within the scope of the present disclosure.

[0230] FIG. 4D is a circuit diagram of an integrated circuit 400D, in accordance with some embodiments.

[0231] Integrated circuit 400D is an embodiment of at least ESD clamp 120, and similar detailed description is therefore omitted.

[0232] Integrated circuit 400D is a variation of integrated circuit 400A of FIG. 4A, and similar detailed description is therefore omitted. In comparison with integrated circuit 400A, ESD assist circuit 404d of integrated circuit 400D replaces ESD assist circuit 404a of integrated circuit 400A, and similar detailed description is therefore omitted.

[0233] Integrated circuit 400D includes control circuit 201, ESD detection circuit 402, ESD assist circuit 404d and discharging circuit 410.

[0234] ESD assist circuit 404d is a variation of ESD assist circuit 404a of FIG. 4A, and similar detailed description is therefore omitted. In comparison with ESD assist circuit 404a, a PMOS transistor P4 of ESD assist circuit 404d replaces NMOS transistor N4 of ESD assist circuit 404a, and similar detailed description is therefore omitted.

[0235] ESD assist circuit 404d includes PMOS transistor P4. PMOS transistor P4 includes a gate, a drain, a source and a body (not labelled).

[0236] In comparison with ESD assist circuit 404a of FIG. 4A, each of the gate of PMOS transistor P4, the source of PMOS transistor P4, the body of PMOS transistor P4, node Nd3, the drain of NMOS transistor N1, the drain of PMOS transistor P1, and the gate of NMOS transistor N3 are coupled together.

[0237] In comparison with ESD assist circuit 404a of FIG. 4A, each of the drain of PMOS transistor P4, the first end of capacitor C2, node Nd2, the source of NMOS transistor N1, the body of NMOS transistor N1, the source of NMOS transistor N3, and the body of NMOS transistor N3 are coupled together.

[0238] The description for when an ESD event occurs at node Nd2 (e.g., ESD current I1 in the forward ESD direction) with ESD assist circuit 404d for FIG. 4D is similar to the description of when an ESD event occurs at node Nd2 for ESD assist circuit 404a of FIG. 4A, and similar detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), PMOS transistor P4 turns on, and is configured to clamp the voltage at node Nd3 to be equal to a voltage of node Nd2 (e.g., VSS) thereby causing the NMOS transistor N3 to turn-on. In response to being turned on,

NMOS transistor N3 couples node Nd2 to node Nd1 and NMOS transistor N3 discharges the ESD current I1 in the forward ESD direction from node Nd2 to Nd1.

[0239] In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), a body diode of PMOS transistor P4 is configured to provide ESD protection of integrated circuit 400D by clamping the voltage of node Nd3. In some embodiments, a body diode of NMOS transistor N3 is configured to provide ESD protection of integrated circuit 400D by passing the ESD current I1 in the forward ESD direction from node Nd2 to node Nd1.

[0240] In some embodiments, by including ESD assist circuit 404d in integrated circuit 400D, integrated circuit 400D achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 4A.

[0241] Other types of circuits, configurations and arrangements of at least ESD detection circuit 402, ESD assist circuit 404d or discharging circuit 410 are within the scope of the present disclosure.

[0242] Other configurations or quantities of circuits in integrated circuit 400D are within the scope of the present disclosure.

[0243] FIG. 4E is a circuit diagram of an integrated circuit 400E, in accordance with some embodiments.

[0244] Integrated circuit 400E is an embodiment of at least ESD clamp 120, and similar detailed description is therefore omitted.

[0245] Integrated circuit 400E is a variation of integrated circuit 400A of FIG. 4A, and similar circuit 404e of integrated circuit 400E replaces ESD assist circuit 404a of integrated circuit 400A, and similar detailed description is therefore omitted.

[0246] Integrated circuit 400E includes control circuit 201, ESD detection circuit 402, ESD assist circuit 404e and discharging circuit 410.

[0247] ESD assist circuit 404e is a variation of ESD assist circuit 404a of FIG. 4A, and similar detailed description is therefore omitted. In comparison with ESD assist circuit 404a, a transistor B4 of ESD assist circuit 404e replaces NMOS transistor N4 of ESD assist circuit 404a, and similar detailed description is therefore omitted. Transistor B4 is a PNP BJT.

[0248] ESD assist circuit 404e includes transistor B4. Transistor B4 includes a base, a collector and an emitter (not labelled).

[0249] In comparison with ESD assist circuit 404a of FIG. 4A, each of the emitter of transistor B4, the base of transistor B4, node Nd3, the drain of NMOS transistor N1, the drain of PMOS transistor P1, and the gate of NMOS transistor N3 are coupled together.

[0250] In comparison with ESD assist circuit 404a of FIG. 4A, each of the collector of transistor B4, the first end of capacitor C2, node Nd2, the source of NMOS transistor N1, the body of NMOS transistor N1, the source of NMOS transistor N3, and the body of NMOS transistor N3 are coupled together.

[0251] The description for when an ESD event occurs at node Nd2 (e.g., ESD current I1 in the forward ESD direction) with ESD assist circuit 404e for FIG. 4E is similar to the description of when an ESD event occurs at node Nd2 for ESD assist circuit 404a of FIG. 4A, and similar detailed description is therefore omitted for brevity. In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), transistor B4 turns on, and is configured to clamp the voltage at node Nd3 to be equal to a voltage of node Nd2 (e.g., VSS) thereby causing the NMOS transistor N3 to turn-on. In response to being turned on, NMOS transistor N3 couples node Nd2 to node Nd1 and NMOS transistor N3 discharges the ESD current I1 in the forward ESD direction from node Nd2 to Nd1.

[0252] In some embodiments, in response to an ESD event at node Nd2 (e.g., an ESD current I1 in the forward ESD direction), a body diode of transistor B4 is configured to provide ESD protection of integrated circuit 400E by clamping the voltage of node Nd3. In some embodiments, a body diode of NMOS transistor N3 is configured to provide ESD protection of integrated circuit 400E by

passing the ESD current I1 in the forward ESD direction from node Nd2 to node Nd1.

[0253] In some embodiments, by including ESD assist circuit 404e in integrated circuit 400E, integrated circuit 400E achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 4A.

[0254] Other types of circuits, configurations and arrangements of at least ESD detection circuit 402, ESD assist circuit 404e or discharging circuit 410 are within the scope of the present disclosure.

[0255] Other configurations or quantities of circuits in integrated circuit 400E are within the scope of the present disclosure.

[0256] FIG. 5A is a cross-sectional view of an integrated circuit 500A, in accordance with some embodiments.

[0257] Integrated circuit 500A is an embodiment of NMOS transistor N4 of ESD assist circuit 404a of FIG. 4A, and similar detailed description is therefore omitted. Integrated circuit 500A is a variation of integrated circuit 300D of FIG. 3D, and similar detailed description is therefore omitted.

[0258] In comparison with integrated circuit 300D of FIG. 3D, a drain region 512b replaces drain region 312b of FIG. 3D, a source region 514b replaces source region 314b of FIG. 3D, tap regions 516b and 517b replace corresponding tap regions 316b and 317b of FIG. 3D, transistor 560b replaces transistor 360b, and similar detailed description is therefore omitted.

[0259] Integrated circuit 500A includes substrate 302a, gate structure 330, drain region 512b, source region 514b, LDD regions 318, sidewalls 320, tap regions 516b and 517b, STI regions 308, 309, 310 and 311, and body diode 540a.

[0260] In FIG. 5A, substrate 302a is a p-type substrate having P-type dopants.

[0261] In some embodiments, drain region 512b, source region 514b, LDD regions 318, sidewalls 320 and gate structure 330 together form transistor 560b. Transistor 560b is an NMOS transistor.

[0262] Drain region 512b corresponds to the drain of NMOS transistor N4 of FIG. 4A, source region 514b corresponds to the source of NMOS transistor N4 of FIG. 4A, gate structure 330 corresponds to the gate of NMOS transistor N4 of FIG. 4A, and tap regions 516b and 517b correspond to the body of NMOS transistor N4 of FIG. 4A, and similar detailed description is therefore omitted.

[0263] The drain region 512b is coupled to node Nd3. The gate structure 330, the source region 514b and the tap regions 516b and 517b are coupled to the reference supply voltage (e.g., voltage VSS) by node Nd2. In some embodiments, each of the gate structure 330, source region 514a and tap regions 516a and 516b are coupled together at node Nd2, and are further coupled to the reference supply voltage (e.g., voltage VSS).

[0264] Integrated circuit 500A further includes a body diode 540a. Body diode 540a is a parasitic element formed by the PN junction between at least tap region 516b or 517b and the drain region 512b. In some embodiments, the body diode 540a is formed by the PN junction between the source region 514b and the drain region 512b. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode 540a of NMOS transistor N4 is configured to provide ESD protection of integrated circuit 500A by clamping the voltage of node Nd3, and integrated circuit 500A thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 4A.

[0265] FIG. 5B is a cross-sectional view of an integrated circuit 500B, in accordance with some embodiments.

[0266] Integrated circuit 500B is an embodiment of diode D2 of ESD assist circuit 404b of FIG. 4B, and similar detailed description is therefore omitted. Integrated circuit 500B is a variation of integrated circuit 300B of FIG. 3B, and similar detailed description is therefore omitted.

[0267] Integrated circuit 500B includes substrate 302a, an N-type region 552a, a P-type region 556a, and STI regions 308, 310 and 311.

[0268] N-type region **552a** is similar to the N-type region **356a** in FIG. 3B, P-type region **556a** is similar to the N-type region **352a** in FIG. 3B, and similar detailed description is therefore omitted.

[0269] The N-type region **552a** is an N+ active region having N-type dopants implanted in the well (not shown) or substrate **302a**. The P-type region **556a** is a P+ active region having P-type dopants implanted in the well (not shown) or substrate **302a**.

[0270] N-type region **552a** corresponds to the cathode of diode D2 of FIG. 4B, and P-type region **556a** corresponds to the anode of diode D2 of FIG. 4B, and similar detailed description is therefore omitted.

[0271] The N-type region **552a** is coupled to node Nd3 (shown in FIG. 2A). The P-type region **556a** is coupled to the supply voltage (e.g., voltage VDD) by node Nd2.

[0272] Integrated circuit **500B** further includes a body diode **540b**. Body diode **540b** is a parasitic element formed by the PN junction between the P-type region **556a** and the N-type region **552a**. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode **540b** is configured to provide ESD protection of integrated circuit **500B** by clamping the voltage of node Nd3, and integrated circuit **500B** thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 4A.

[0273] FIG. 5C is a cross-sectional view of an integrated circuit **500C**, in accordance with some embodiments.

[0274] Integrated circuit **500C** is an embodiment of transistor B3 of ESD assist circuit **404c** of FIG. 4C, and similar detailed description is therefore omitted. Integrated circuit **500C** is a variation of integrated circuit **300E** of FIG. 3E, and similar detailed description is therefore omitted.

[0275] Integrated circuit **500C** includes substrate **302b**, N-type region **362b**, and P-type region **366b**.

[0276] N-type region **362b** corresponds to the emitter of transistor B3 of FIG. 4C, P-type region **366b** corresponds to the base of transistor B3 of FIG. 4C, substrate **302b** corresponds to the collector of transistor B3 of FIG. 4C, and similar detailed description is therefore omitted.

[0277] Each of the N-type region **362b** and the P-type region **366b** are coupled together, and further coupled to the reference supply voltage (e.g., voltage VSS) by node Nd2. The substrate **302b** is coupled to node Nd3.

[0278] Integrated circuit **500C** further includes a body diode **540c**. Body diode **540c** is a parasitic element formed by the PN junction between the P-type region **366b** and the substrate **302b**. In some embodiments, in response to ESD events at node Nd2 in the forward direction, body diode **540c** is configured to provide ESD protection of integrated circuit **500C** by clamping the voltage of node Nd3, and integrated circuit **500C** thereby achieves one or more of the benefits discussed above in at least FIG. 1 or FIG. 4A.

[0279] FIG. 5D is a cross-sectional view of an integrated circuit **500D**, in accordance with some embodiments.

[0280] Integrated circuit **500D** is an embodiment of PMOS transistor P4 of ESD assist circuit **404d** of FIG. 4D, and similar detailed description is therefore omitted. Integrated circuit **500D** is a variation of integrated circuit **300A** of FIG. 3A, and similar detailed description is therefore omitted.

[0281] In comparison with integrated circuit **300A** of FIG. 3A, a drain region **512a** replaces drain region **312a** of FIG. 3D, a source region **514a** replaces source region **314a** of FIG. 3D, tap regions **516a** and **517a** replace corresponding tap regions **316a** and **317a** of FIG. 3D, transistor **560a** replaces transistor **360a**, and similar detailed description is therefore omitted.

[0282] Integrated circuit **500D** includes substrate **302b**, gate structure **330**, drain region **512a**, source region **514a**, LDD regions **318**, sidewalls **320**, tap regions **516a** and **517a**, STI regions **308**, **309**, **310** and **311**, and body diode **540d**.

[0283] In FIG. 5D, substrate **302b** is an n-type substrate having N-type dopants.

[0284] In some embodiments, drain region **512a**, source region **514a**, LDD regions **318**, sidewalls

320 and gate structure **330** together form transistor **560a**. Transistor **560a** is a PMOS transistor. [0285] Drain region **512a** corresponds to the drain of PMOS transistor **P4** of FIG. **4A**, source region **514a** corresponds to the source of PMOS transistor **P4** of FIG. **4A**, gate structure **330** corresponds to the gate of PMOS transistor **P4** of FIG. **4A**, and tap regions **516a** and **517a** correspond to the body of PMOS transistor **P4** of FIG. **4A**, and similar detailed description is therefore omitted.

[0286] The drain region **512a** is coupled to the reference supply voltage (e.g., voltage **VSS**) by node **Nd2**. The gate structure **330**, the source region **514a** and the tap regions **516a** and **517a** are coupled to node **Nd3**.

[0287] Integrated circuit **500D** further includes a body diode **540d**. Body diode **540d** is a parasitic element formed by the PN junction between drain region **512a** and at least tap region **516a** or **517a**. In some embodiments, the body diode **540d** is formed by the PN junction between the drain region **512a** and the source region **514a**. In some embodiments, in response to ESD events at node **Nd2** in the forward direction, body diode **540d** of PMOS transistor **P4** is configured to provide ESD protection of integrated circuit **500D** by clamping the voltage of node **Nd3**, and integrated circuit **500D** thereby achieves one or more of the benefits discussed above in at least FIG. **1** or FIG. **4A**.

[0288] FIG. **5E** is a cross-sectional view of an integrated circuit **500E**, in accordance with some embodiments.

[0289] Integrated circuit **500E** is an embodiment of transistor **B4** of ESD assist circuit **404e** of FIG. **4E**, and similar detailed description is therefore omitted. Integrated circuit **500E** is a variation of integrated circuit **300C** of FIG. **3C**, and similar detailed description is therefore omitted.

[0290] Integrated circuit **500E** includes substrate **302a**, P-type region **362a**, and N-type region **366a**.

[0291] In FIG. **5E**, substrate **302a** is a p-type substrate having P-type dopants.

[0292] P-type region **362a** corresponds to the emitter of transistor **B4** of FIG. **4E**, N-type region **366a** corresponds to the base of transistor **B4** of FIG. **4E**, substrate **302a** corresponds to the collector of transistor **B4** of FIG. **4E**, and similar detailed description is therefore omitted.

[0293] Each of the P-type region **362a** and the N-type region **366a** are coupled together, and are further coupled to node **Nd3**. The substrate **302a** is coupled to the reference supply voltage (e.g., voltage **VSS**) by node **Nd2**.

[0294] Integrated circuit **500E** further includes a body diode **540e**. Body diode **540e** is a parasitic element formed by the PN junction between the substrate **302a** and the N-type region **366a**. In some embodiments, in response to ESD events at node **Nd2** in the forward direction, body diode **540e** is configured to provide ESD protection of integrated circuit **500E** by clamping the voltage of node **Nd3**, and integrated circuit **500E** thereby achieves one or more of the benefits discussed above in at least FIG. **1** or FIG. **4A**.

[0295] FIG. **6A** is a circuit diagram of an integrated circuit **600A**, in accordance with some embodiments.

[0296] Integrated circuit **600A** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted. In some embodiments, integrated circuit **600A** is a 2-stacked P-type power clamp circuit.

[0297] Integrated circuit **600A** includes circuits **601a** and **601b** arranged in a stacked configuration. In some embodiments, each of circuit **601a** and circuit **601b** corresponds to integrated circuit **200A**, **200B**, **200C**, **200D** or **200E** arranged in a stacked configuration, and similar detailed description is therefore omitted. In some embodiments, a voltage of an ESD event is larger than a withstanding voltage of a single ESD clamp circuit, so one or more ESD clamp circuits are arranged in a stacked configuration to withstand ESD events with a voltage greater than the withstanding voltage of a single ESD clamp circuit, but less than the total withstanding voltage of a plurality of ESD clamp circuits thereby preventing failure during the ESD event.

[0298] Circuit **601a** is coupled to circuit **601b** at node **Nd2**.

[0299] Circuit **601a** includes ESD detection circuit **602a**, ESD assist circuit **604a** and discharging circuit **610a**.

[0300] Circuit **601b** includes ESD detection circuit **602b**, ESD assist circuit **604b** and discharging circuit **610b**.

[0301] Each of ESD detection circuit **601a** and **601b** corresponds to ESD detection circuit **202** of FIGS. 2A-2E, each of ESD assist circuit **604a** and **604b** corresponds to at least ESD assist circuit **204a** of FIG. 2A, **204b** of FIG. 2B, **204c** of FIG. 2C, **204d** of FIG. 2D or **204e** of FIG. 2E, each of ESD discharge circuit **610a** and **610b** corresponds to ESD discharge circuit **210** of FIGS. 2A-2E, and similar detailed description is therefore omitted.

[0302] Circuit **601a** is coupled between nodes Nd1 and Nd2.

[0303] Circuit **601b** is coupled between nodes Nd2 and Nd5.

[0304] ESD detection circuit **602a** is coupled between nodes Nd1 and Nd2. ESD detection circuit **602b** is coupled between nodes Nd2 and Nd5. In comparison with FIGS. 2A-2E, node Nd1 has the supply voltage VDD, and node Nd5 has the reference voltage VSS.

[0305] ESD discharge circuits **610a** and **610b** are coupled in series across node Nd1 and node Nd5. ESD discharge circuit **610a** is coupled between nodes Nd1 and Nd2. ESD discharge circuit **610b** is coupled between nodes Nd2 and Nd5.

[0306] ESD assist circuit **604a** is coupled between nodes Nd1 and Nd3. ESD assist circuit **604b** is coupled between nodes Nd2 and Nd6. Nodes Nd6 and Nd5 in FIGS. 6A-6B are similar to corresponding nodes Nd3 and Nd2 in FIGS. 2A-2E, and similar detailed description is therefore omitted. In comparison with FIGS. 2A-2E, a voltage of node Nd5 in FIGS. 6A & 7A (described below) corresponds to reference supply voltage VSS, and a voltage of node Nd2 in FIGS. 6A & 7A corresponds to intermediate voltage V1a. In some embodiments, intermediate voltage V1a is equal to $(VDD-VSS)/2$.

[0307] Capacitive coupling effects between node Nd2 and Nd3 are shown by equivalent capacitor C1a. In some embodiments, by including ESD assist circuits **604a** and **604b** in integrated circuit **600A**, capacitive coupling effects (shown by equivalent capacitor C1a) between node Nd2 and Nd3 are reduced thereby effectively lowering the turn-on resistance of ESD discharge circuits **610a** and **610b**, and thus causing the ESD discharge circuits **610a** and **610b** to turn-on faster and stronger than other approaches resulting in improved ESD performance and improved ESD robustness compared to other approaches.

[0308] Other types of circuits, configurations and arrangements of at least ESD detection circuits **602a** and **602b**, ESD assist circuits **604a** and **604b** or discharging circuits **610a** and **610b** are within the scope of the present disclosure.

[0309] Other configurations or quantities of circuits in integrated circuit **600A** are within the scope of the present disclosure.

[0310] FIG. 6B is a circuit diagram of an integrated circuit **600B**, in accordance with some embodiments.

[0311] Integrated circuit **600B** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted. In some embodiments, integrated circuit **600B** is a 3-stacked P-type power clamp circuit.

[0312] Integrated circuit **600B** is a variation of integrated circuit **600A** of FIG. 6A, and similar detailed description is therefore omitted. In comparison with integrated circuit **600A**, integrated circuit **600B** further includes circuit **601c**, and similar detailed description is therefore omitted. Circuit **601c** is coupled to circuit **601b** at node Nd2.

[0313] Circuit **601c** includes ESD detection circuit **602c**, ESD assist circuit **604c** and discharging circuit **610c**.

[0314] ESD detection circuit **601c** corresponds to ESD detection circuit **202** of FIGS. 2A-2E, ESD assist circuit **604c** corresponds to at least ESD assist circuit **204a** of FIG. 2A, **204b** of FIG. 2B, **204c** of FIG. 2C, **204d** of FIG. 2D or **204e** of FIG. 2E, ESD discharge circuit **610c** corresponds to

ESD discharge circuit **210** of FIGS. 2A-2E, and similar detailed description is therefore omitted. Circuit **601c** is coupled between nodes Nd5 and Nd7.

[0315] ESD detection circuit **602c** is coupled between nodes Nd5 and Nd7. In comparison with FIG. 6A, a voltage of node Nd7 corresponds to reference supply voltage VSS, a voltage of node Nd2 in FIGS. 6B & 7B (described below) corresponds to intermediate voltage V1b, and a voltage of node Nd5 in FIGS. 6B & 7B corresponds to intermediate voltage V2b. In some embodiments, intermediate voltage V2b is equal to $(VDD-VSS)/3$. In some embodiments, intermediate voltage V1b is equal to $2*(VDD-VSS)/3$.

[0316] ESD discharge circuits **610a**, **610b** and **610c** are coupled in series across node Nd1 and node Nd7. ESD discharge circuit **610c** is coupled between nodes Nd5 and Nd7.

[0317] ESD assist circuit **604c** is coupled between nodes Nd5 and Nd8. Nodes Nd5 and Nd8 in FIGS. 6B & 7B are similar to corresponding nodes Nd3 and Nd2 in FIGS. 2A-2E, and similar detailed description is therefore omitted.

[0318] In some embodiments, by including ESD assist circuits **604a**, **604b** and **604c** in integrated circuit **600B**, capacitive coupling effects (shown by equivalent capacitor C1a) between node Nd2 and Nd3 are reduced thereby effectively lowering the turn-on resistance of ESD discharge circuits **610a**, **610b** and **610c**, and thus causing the ESD discharge circuits **610a**, **610b** and **610c** to turn-on faster and stronger than other approaches resulting in improved ESD performance and improved ESD robustness compared to other approaches.

[0319] Other types of circuits, configurations and arrangements of at least ESD detection circuits **602a**, **602b** and **602c**, ESD assist circuits **604a**, **604b** and **604c** discharging circuits **610a**, **610b** and **610c** are within the scope of the present disclosure.

[0320] Other configurations or quantities of circuits in integrated circuit **600B** are within the scope of the present disclosure. For example, in some embodiments, integrated circuit **600B** includes other numbers of circuit **601a**, **601b** or **601c** (e.g., 3-stacked P-type power clamp circuit) than 3.

[0321] FIG. 7A is a circuit diagram of an integrated circuit **700A**, in accordance with some embodiments.

[0322] Integrated circuit **700A** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted. In some embodiments, integrated circuit **700A** is a 2-stacked N-type power clamp circuit.

[0323] Integrated circuit **700A** is a variation of integrated circuit **600A** of FIG. 6A, and similar detailed description is therefore omitted. In comparison with integrated circuit **600A**, integrated circuit **700A** includes features of integrated circuit **400A-400E** of FIGS. 4A-4E rather than integrated circuit **200A-200E** of FIGS. 2A-2E, and similar detailed description is therefore omitted.

[0324] Integrated circuit **700A** includes circuits **701a** and **701b** arranged in a stacked configuration. In some embodiments, each of circuit **701a** and circuit **701b** corresponds to integrated circuit **400A**, **400B**, **400C**, **400D** or **400E** arranged in a stacked configuration, and similar detailed description is therefore omitted.

[0325] Circuit **701a** is coupled to circuit **701b** at node Nd2.

[0326] Circuit **701a** includes ESD detection circuit **702a**, ESD assist circuit **704a** and discharging circuit **710a**.

[0327] Circuit **701b** includes ESD detection circuit **702b**, ESD assist circuit **704b** and discharging circuit **710b**.

[0328] Each of ESD detection circuit **701a** and **701b** corresponds to ESD detection circuit **402** of FIGS. 4A-4E, each of ESD assist circuit **704a** and **704b** corresponds to at least ESD assist circuit **404a** of FIG. 4A, **404b** of FIG. 4B, **404c** of FIG. 4C, **404d** of FIG. 4D or **404e** of FIG. 4E, each of ESD discharge circuit **710a** and **710b** corresponds to ESD discharge circuit **410** of FIGS. 4A-4E, and similar detailed description is therefore omitted.

[0329] Circuit **701a** is coupled between nodes Nd1 and Nd2.

[0330] Circuit **701b** is coupled between nodes Nd2 and Nd5.

[0331] ESD detection circuit **702a** is coupled between nodes Nd1 and Nd2. ESD detection circuit **702b** is coupled between nodes Nd2 and Nd5. In comparison with FIGS. 4A-4E, node Nd1 has the supply voltage VDD, and node Nd5 has the reference voltage VSS.

[0332] ESD discharge circuits **710a** and **710b** are coupled in series across node Nd1 and node Nd5. ESD discharge circuit **710a** is coupled between nodes Nd1 and Nd2. ESD discharge circuit **710b** is coupled between nodes Nd2 and Nd5.

[0333] ESD assist circuit **704a** is coupled between nodes Nd2 and Nd3. ESD assist circuit **704b** is coupled between nodes Nd5 and Nd6.

[0334] Capacitive coupling effects between node Nd1 and Nd3 are shown by equivalent capacitor C1b. In some embodiments, by including ESD assist circuits **704a** and **704b** in integrated circuit **700A**, capacitive coupling effects (shown by equivalent capacitor C1b) between node Nd1 and Nd3 are reduced thereby effectively lowering the turn-on resistance of ESD discharge circuits **710a** and **710b**, and thus causing the ESD discharge circuits **710a** and **710b** to turn-on faster and stronger than other approaches resulting in improved ESD performance and improved ESD robustness compared to other approaches.

[0335] Other types of circuits, configurations and arrangements of at least ESD detection circuits **702a** and **702b**, ESD assist circuits **704a** and **704b** or discharging circuits **710a** and **710b** are within the scope of the present disclosure.

[0336] Other configurations or quantities of circuits in integrated circuit **700A** are within the scope of the present disclosure.

[0337] FIG. 7B is a circuit diagram of an integrated circuit **700B**, in accordance with some embodiments.

[0338] Integrated circuit **700B** is an embodiment of at least ESD clamp **120**, and similar detailed description is therefore omitted. In some embodiments, integrated circuit **700B** is a 3-stacked N-type power clamp circuit.

[0339] Integrated circuit **700B** is a variation of integrated circuit **600B** of FIG. 6B and integrated circuit **700A** of FIG. 7A, and similar detailed description is therefore omitted. In comparison with integrated circuit **600B**, integrated circuit **700B** includes features of integrated circuit **400A-400E** of FIGS. 4A-4E rather than integrated circuit **200A-200E** of FIGS. 2A-2E, and similar detailed description is therefore omitted. In comparison with integrated circuit **700A**, integrated circuit **700B** further includes circuit **701c**, and similar detailed description is therefore omitted.

[0340] Circuit **701c** is coupled to circuit **701b** at node Nd2.

[0341] Circuit **701c** includes ESD detection circuit **702c**, ESD assist circuit **704c** and discharging circuit **710c**.

[0342] ESD detection circuit **701c** corresponds to ESD detection circuit **402** of FIGS. 4A-4E, ESD assist circuit **704c** corresponds to at least ESD assist circuit **404a** of FIG. 4A, **404b** of FIG. 4B, **404c** of FIG. 4C, **404d** of FIG. 4D or **404e** of FIG. 4E, ESD discharge circuit **710c** corresponds to ESD discharge circuit **410** of FIGS. 4A-4E, and similar detailed description is therefore omitted.

[0343] Circuit **701c** is coupled between nodes Nd5 and Nd7.

[0344] ESD detection circuit **702c** is coupled between nodes Nd5 and Nd7. In comparison with FIG. 7A, the voltage of node Nd7 corresponds to reference supply voltage VSS.

[0345] ESD discharge circuits **710a**, **710b** and **710c** are coupled in series across node Nd1 and node Nd7. ESD discharge circuit **710c** is coupled between nodes Nd5 and Nd7.

[0346] ESD assist circuit **704c** is coupled between nodes Nd7 and Nd8. Nodes Nd7 and Nd8 in FIGS. 6B & 7B are similar to corresponding nodes Nd2 and Nd3 in FIGS. 4A-4E, and similar detailed description is therefore omitted.

[0347] In some embodiments, by including ESD assist circuits **704a**, **704b** and **704c** in integrated circuit **700B**, capacitive coupling effects (shown by equivalent capacitor C1b) between node Nd1 and Nd3 are reduced thereby effectively lowering the turn-on resistance of ESD discharge circuits **710a**, **710b** and **710c**, and thus causing the ESD discharge circuits **710a**, **710b** and **710c** to turn-on

faster and stronger than other approaches resulting in improved ESD performance and improved ESD robustness compared to other approaches.

[0348] Other types of circuits, configurations and arrangements of at least ESD detection circuits **702a**, **702b** and **702c**, ESD assist circuits **704a**, **704b** and **704c** discharging circuits **710a**, **710b** and **710c** are within the scope of the present disclosure.

[0349] Other configurations or quantities of circuits in integrated circuit **700B** are within the scope of the present disclosure. For example, in some embodiments, integrated circuit **700B** includes other numbers of circuit **701a**, **701b** or **701c** (e.g., 3-stacked N-type power clamp circuit) than 3.

[0350] FIGS. **8A-8B** are top-views of corresponding integrated circuits **800A-800B**, in accordance with some embodiments.

[0351] Integrated circuits **800A-800B** are embodiments of at least ESD clamp **120**, and similar detailed description is therefore omitted.

[0352] Integrated circuit **800A** is a top-level view of at least integrated circuit **200A**, **200B**, **200C**, **200D** or **200E**, in accordance with some embodiments.

[0353] Integrated circuit **800B** is a top-level view of at least integrated circuit **400A**, **400B**, **400C**, **400D** or **400E**, in accordance with some embodiments.

[0354] Integrated circuit **800A** or **800B** is manufactured by a corresponding layout design similar to corresponding integrated circuit **800A** or **800B**. For brevity FIGS. **8A-8B** are described as integrated circuits **800A-800B**, but in some embodiments, FIGS. **8A-8B** also correspond to layout designs, and structural elements of integrated circuits **800A-800B** also correspond to layout patterns, and structural relationships including alignment, lengths and widths, as well as configurations and layers of a corresponding layout design of integrated circuits **800A-800B** are similar to the structural relationships and configurations and layers of integrated circuits **800A-800B**, and similar detailed description will not be described for brevity.

[0355] Integrated circuit **800A** includes regions **801**, **802a**, **802b**, **804a** and **810a**, conductive structures **820a**, **822a**, **824a**, **826a**, **828a**, **830a** and **840a**.

[0356] Region **801** corresponds to control circuit **201** of FIGS. **2A-2E** & **4A-4E**, region **802a** corresponds to PMOS transistor **P1** of ESD detection circuit **202**, region **802b** corresponds to NMOS transistor **N1** of ESD detection circuit **202**, region **804a** corresponds to at least ESD assist circuit **204a** of FIG. **2A**, **204b** of FIG. **2B**, **204c** of FIG. **2C**, **204d** of FIG. **2D** or **204e** of FIG. **2E**, region **810a** corresponds to ESD discharge circuit **210** of FIGS. **2A-2E**, and similar detailed description is therefore omitted.

[0357] Conductive structure **822a** or conductive structure **828a** corresponds to node **Nd2** of at least ESD assist circuit **204a** of FIG. **2A**, **204b** of FIG. **2B**, **204c** of FIG. **2C**, **204d** of FIG. **2D** or **204e** of FIG. **2E**, and similar detailed description is therefore omitted.

[0358] Conductive structure **824a** or conductive structure **826a** corresponds to node **Nd1** of at least ESD assist circuit **204a** of FIG. **2A**, **204b** of FIG. **2B**, **204c** of FIG. **2C**, **204d** of FIG. **2D** or **204e** of FIG. **2E**, and similar detailed description is therefore omitted.

[0359] Conductive structure **840a** corresponds to node **Nd3** of at least ESD assist circuit **204a** of FIG. **2A**, **204b** of FIG. **2B**, **204c** of FIG. **2C**, **204d** of FIG. **2D** or **204e** of FIG. **2E**, and similar detailed description is therefore omitted.

[0360] At least conductive structure **820a**, **824a**, **826a** or **830a** is configured to supply a supply voltage **VDD**. In some embodiments, at least conductive structure **820a**, **824a**, **826a** or **830a** is a power rail.

[0361] At least conductive structure **822a** or **828a** is configured to supply a reference supply voltage **VSS**. In some embodiments, at least conductive structure **822a** or **828a** is a reference power rail.

[0362] Each of conductive structures **820a**, **822a**, **824a**, **826a**, **828a**, **830a** and **840a** extend in a first direction **X**, and are separated from each other in a second direction **Y** different from the first direction **X**. In some embodiments, each of conductive structures **820a**, **822a**, **824a**, **826a**, **828a**,

830a and **840a** are on a same level of integrated circuit **800A**. In some embodiments, at least one of conductive structure **820a**, **822a**, **824a**, **826a**, **828a**, **830a** or **840a** is on a different level of integrated circuit **800A** from another of at least one of conductive structure **820a**, **822a**, **824a**, **826a**, **828a**, **830a** or **840a**.

[0363] In some embodiments, at least one conductive structure of conductive structures **820a**, **822a**, **824a**, **826a**, **828a**, **830a** and **840a** includes one or more layers of a conductive material, a metal, a metal compound or a doped semiconductor. In some embodiments, the conductive material includes Tungsten, Cobalt, Ruthenium, Copper, or the like or combinations thereof. In some embodiments, a metal includes at least Cu (Copper), Co, W, Ru, Al, or the like. In some embodiments, a metal compound includes at least AlCu, W-TiN, TiSix, NiSix, TiN, TaN, or the like. In some embodiments, a doped semiconductor includes at least doped silicon, or the like.

[0364] Conductive structure **824a** is positioned between conductive structure **804a** and conductive structure **822a**. Conductive structures **804a** and **822a** are separated from each other by a distance **D1** in the second direction **Y**.

[0365] Conductive structure **826a** is positioned between conductive structure **804a** and conductive structure **828a**. Conductive structures **804a** and **828a** are separated from each other by distance **D1** in the second direction **Y**.

[0366] In some embodiments, conductive structure **824a** is positioned between conductive structure **804a** and conductive structure **822a**, which causes conductive structure **804a** and conductive structure **822a** to be separated from each other in the second direction **Y** by distance **D1** that is greater than other approaches, thereby results in less coupled capacitance between conductive structure **804a** and conductive structure **822a** than other approaches.

[0367] In some embodiments, conductive structure **826a** is positioned between conductive structure **804a** and conductive structure **828a**, which causes conductive structure **804a** and conductive structure **828a** to be separated from each other in the second direction **Y** by distance **D1** that is greater than other approaches, thereby results in less coupled capacitance between conductive structure **804a** and conductive structure **828a** than other approaches.

[0368] Other configurations or quantities of circuits in integrated circuit **800A** are within the scope of the present disclosure.

[0369] Integrated circuit **800B** is a top-level view of at least integrated circuit **400A**, **400B**, **400C**, **400D** or **400E**, in accordance with some embodiments.

[0370] Integrated circuit **800B** is a variation of integrated circuit **800A** of FIG. **8A**, and similar detailed description is therefore omitted. In comparison with integrated circuit **800A**, region **804b** of FIG. **8B** replaces region **804a**, region **810b** of FIG. **8B** replaces region **810a**, conductive structures **820b**, **822b**, **824b**, **826b**, **828b**, **830b** and **840b** replace corresponding conductive structures **820a**, **822a**, **824a**, **826a**, **828a**, **830a** and **840a**, and similar detailed description is therefore omitted.

[0371] Integrated circuit **800B** includes regions **801**, **802a**, **802b**, **804b** and **810b**, conductive structures **820b**, **822b**, **824b**, **826b**, **828b**, **830b** and **840b**.

[0372] Region **804b** corresponds to at least ESD assist circuit **404a**, **404b**, **404c**, **404d** or **404e** of corresponding FIGS. **4A-4E**, region **810b** corresponds to ESD discharge circuit **410** of FIGS. **4A-4E**, and similar detailed description is therefore omitted.

[0373] Conductive structure **822b** or conductive structure **828b** corresponds to node **Nd1** of at least ESD assist circuit **404a**, **404b**, **404c**, **404d** or **404e** of corresponding FIGS. **4A-4E**, and similar detailed description is therefore omitted.

[0374] Conductive structure **824b** or conductive structure **826b** corresponds to node **Nd2** of at least ESD assist circuit **404a**, **404b**, **404c**, **404d** or **404e** of corresponding FIGS. **4A-4E**, and similar detailed description is therefore omitted.

[0375] Conductive structure **840b** corresponds to node **Nd3** of at least ESD assist circuit **404a**, **404b**, **404c**, **404d** or **404e** of corresponding FIGS. **4A-4E**, and similar detailed description is

therefore omitted.

[0376] At least conductive structure **820b**, **824b**, **826b** or **830b** is configured to supply the reference supply voltage VSS. In some embodiments, at least conductive structure **820b**, **824b**, **826b** or **830b** is a reference power rail.

[0377] At least conductive structure **822b** or **828b** is configured to supply the supply voltage VDD. In some embodiments, at least conductive structure **822b** or **828b** is a power rail.

[0378] In some embodiments, each of conductive structures **820b**, **822b**, **824b**, **826b**, **828b**, **830b** and **840b** are on a same level of integrated circuit **800B**. In some embodiments, at least one of conductive structure **820b**, **822b**, **824b**, **826b**, **828b**, **830b** or **840b** is on a different level of integrated circuit **800B** from another of at least one of conductive structure **820b**, **822b**, **824b**, **826b**, **828b**, **830b** or **840b**.

[0379] Conductive structure **824b** is positioned between conductive structure **804b** and conductive structure **822b**. Conductive structures **804b** and **822b** are separated from each other by distance D1 in the second direction Y.

[0380] Conductive structure **826b** is positioned between conductive structure **804b** and conductive structure **828b**. Conductive structures **804b** and **828b** are separated from each other by distance D1 in the second direction Y.

[0381] In some embodiments, conductive structure **824b** is positioned between conductive structure **804b** and conductive structure **822b**, which causes conductive structure **804b** and conductive structure **822b** to be separated from each other in the second direction Y by distance D1 that is greater than other approaches, thereby results in less coupled capacitance between conductive structure **804b** and conductive structure **822b** than other approaches.

[0382] In some embodiments, conductive structure **826b** is positioned between conductive structure **804b** and conductive structure **828b**, which causes conductive structure **804b** and conductive structure **828b** to be separated from each other in the second direction Y by distance D1 that is greater than other approaches, thereby results in less coupled capacitance between conductive structure **804b** and conductive structure **828b** than other approaches.

[0383] Other configurations or quantities of circuits in integrated circuit **800B** are within the scope of the present disclosure.

[0384] FIG. 9 is a flowchart of a method **900** of operating an ESD circuit, in accordance with some embodiments.

[0385] In some embodiments, the ESD circuit of method **900** includes at least integrated circuit **100**, **200A-200E**, **300A-300E**, **400A-400E**, **500A-500E**, **600A-600B**, **700A-700B** and **800A-800B** (FIGS. 1, 2A-2E, 3A-3E, 4A-4E, 5A-5E, 6A-6B, 7A-7B and 8A-8B). It is understood that additional operations may be performed before, during, and/or after the method **900** depicted in FIG. 9, and that some other processes may only be briefly described herein. It is understood that method **900** utilizes features of one or more of integrated circuit **100**, **200A-200E**, **300A-300E**, **400A-400E**, **500A-500E**, **600A-600B**, **700A-700B** and **800A-800B**. Other order of operations of method **900** with respect to at least integrated circuit **100**, **200A-200E**, **300A-300E**, **400A-400E**, **500A-500E**, **600A-600B**, **700A-700B** or **800A-800B** is within the scope of the present disclosure.

[0386] At operation **902** of method **900**, a first ESD voltage is received on a first node. In some embodiments, the first node of method **900** includes node Nd2. In some embodiments, the first ESD voltage is greater than a reference supply voltage VSS of node Nd2. In some embodiments, the first ESD voltage corresponds to a first ESD event.

[0387] At operation **904**, a detection circuit detects the first ESD event at the first node thereby causing the detection circuit to charge a first gate of a first transistor of a discharging circuit.

[0388] In some embodiments, the detection circuit of method **900** includes at least detection circuit **202** or **402**. In some embodiments, the discharging circuit of method **900** includes at least discharging circuit **210** or **410**. In some embodiments, the first transistor of the discharging circuit of method **900** includes at least PMOS transistor P2 or NMOS transistor N3.

[0389] In some embodiments, the discharging circuit is coupled between the first node and a second node. In some embodiments, the detection circuit is coupled to the first node, the second node and a third node.

[0390] In some embodiments, the second node of method **900** includes node Nd1. In some embodiments, the third node of method **900** includes node Nd3 or Nd4.

[0391] At operation **906**, an ESD assist circuit is turned on in response to the first ESD event at the first node thereby causing the ESD assist circuit to clamp a first voltage of the first gate of the first transistor of the discharging circuit at a second voltage of a second node or a third voltage of the first node.

[0392] In some embodiments, the ESD assist circuit of method **900** includes at least ESD assist circuit **204a**, **204b**, **204c**, **204d**, **204e**, **404a**, **404b**, **404c**, **404d** or **404e**. In some embodiments, the ESD assist circuit of method **900** includes at least integrated circuit **300A**, **300B**, **300C**, **300D**, **300E**, **500A**, **500B**, **500C**, **500D** or **500E**.

[0393] In some embodiments, the first voltage of method **900** is the voltage of node Nd3. In some embodiments, the second voltage of the second node of method **900** is the voltage (e.g., VDD) of node Nd1. In some embodiments, the third voltage of the first node of method **900** is the voltage (e.g., VSS) of node Nd2.

[0394] In some embodiments, the ESD assist circuit is coupled between at least the first node and the third node.

[0395] In some embodiments, the ESD assist circuit includes a body diode. In some embodiments, the body diode of method **900** includes at least body diode **340a**, **340b**, **340c**, **340d**, **340e**, **540a**, **540b**, **540c**, **540d** or **540e**.

[0396] At operation **908**, the first transistor of the discharging circuit is turned on in response to at least the first ESD event at the first node and the ESD assist circuit clamping the first voltage of the first gate at the second voltage or the third voltage. In some embodiments, operation **908** further includes the first transistor of the discharging circuit being turned on in response to the first gate of the first transistor being charged.

[0397] At operation **910**, the first node is coupled to the second node in response to the first transistor of the discharging circuit turning on.

[0398] At operation **912** a first ESD current of the first ESD event at the first node is discharged in a first ESD direction from the first node to the second node by the first transistor. In some embodiments, operation **912** further includes the first ESD current of the first ESD event at the first node being discharged in the first ESD direction from the first node to the second node by at least a channel of the first transistor or a body diode of the first transistor.

[0399] In some embodiments, the first ESD current corresponds to the forward ESD direction. In some embodiments, the first ESD current includes the ESD current I1 in the forward ESD direction from node Nd2 to node Nd1.

[0400] In some embodiments, one or more of the operations of method **900** is not performed. In some embodiments, one or more of the operations of method **900** is repeated. In some embodiments, method **900** is repeated.

[0401] In some embodiments, method **900** and **1000** (discussed below) thereby achieves one or more of the benefits discussed above in at least FIG. 1, 2A-2E, 3A-3E, 4A-4E, 5A-5E, 6A-6B, 7A-7B or 8A-8B.

[0402] FIG. 10 is a flow chart of a method **1000** of manufacturing an integrated circuit, in accordance with some embodiments.

[0403] In some embodiments, the method **1000** is usable to manufacture or fabricate at least integrated circuit **100**, **200A-200E**, **300A-300E**, **400A-400E**, **500A-500E**, **600A-600B**, **700A-700B** and **800A-800B** (FIGS. 1, 2A-2E, 3A-3E, 4A-4E, 5A-5E, 6A-6B, 7A-7B and 8A-8B).

[0404] It is understood that additional operations may be performed before, during, and/or after the method **1000** depicted in FIG. 10, and that some other processes may only be briefly described

herein. It is understood that method **1000** utilizes features of one or more of integrated circuit **100**, **200A-200E**, **300A-300E**, **400A-400E**, **500A-500E**, **600A-600B**, **700A-700B** and **800A-800B**.

[0405] Other order of operations of method **1000** with respect to at least integrated circuit **100**, **200A-200E**, **300A-300E**, **400A-400E**, **500A-500E**, **600A-600B**, **700A-700B** or **800A-800B** is within the scope of the present disclosure.

[0406] Method **1000** is applicable to at least integrated circuit **500A**, **500B** or **500C**. Method **1000** is described with respect to integrated circuit **500A**, **500B** or **500C**. However, method **1000** is also applicable to integrated circuit **100-100B**, **200A-200B**, **300A-300B** or **400A-400C**. Other order of operations of method **1000** with respect to integrated circuit **500A**, **500B** or **500C** is within the scope of the present disclosure.

[0407] In operation **1002** of method **1000**, an ESD assist circuit is fabricated on a wafer. In some embodiments, the ESD assist circuit of method **1000** includes at least ESD assist circuit **204a**, **204b**, **204c**, **204d**, **204e**, **404a**, **404b**, **404c**, **404d** or **404e**. In some embodiments, the ESD assist circuit of method **1000** includes at least ESD assist circuit **300A**, **300B**, **300C**, **300D**, **300E**, **500A**, **500B**, **500C**, **500D** or **500E**. In some embodiments, the wafer of method **1000** includes substrate **302a** or **302b**.

[0408] In some embodiments, operation **1002** includes fabricating at least integrated circuit **300A**, **300B**, **300C**, **300D**, **300E**, **500A**, **500B**, **500C**, **500D** or **500E**.

[0409] In some embodiments, operation **1002** includes fabricating at least integrated circuit **300A** or **500D** where a first well is fabricated in a substrate, a drain region and a source region of a transistor is fabricated in the first well, and a first gate structure is fabricated. In these embodiments, the substrate is substrate **302a** which includes p-type dopants, the first well (not shown) includes n-type dopants, the drain region is drain region **312a** or **512a** which includes p-type dopants, and the source region is source region **314a** or **514a** which includes p-type dopants, and the first gate structure is gate **330**. In some embodiments, the p-dopants include boron, aluminum or other suitable p-type dopants.

[0410] In some embodiments, the first well comprises n-type dopants. In some embodiments, the n-dopants include phosphorus, arsenic or other suitable n-type dopants. In some embodiments, the first well comprises an epi-layer grown over substrate **302a** or **302b**. In some embodiments, the epi-layer is doped by adding dopants during the epitaxial process. In some embodiments, the epi-layer is doped by ion implantation after the epi-layer is formed. In some embodiments, the first well is formed by doping substrate **302a** or **302b**. In some embodiments, the doping is performed by ion implantation. In some embodiments, the first well has a dopant concentration ranging from 1×10^{12} atoms/cm³ to 1×10^{14} atoms/cm³. Other values or ranges of dopant concentrations for the first well are within the scope of the present disclosure.

[0411] In some embodiments, the drain region of transistor **360a** or **560a** is fabricated in the first well. In these embodiments, the drain region extends in the second direction Y, and has the p-type dopants. In some embodiments, the source region of transistor **360a** or **560a** is fabricated in the first well. In these embodiments, the source region extends in the second direction Y, and has the p-type dopants.

[0412] In some embodiments, operation **1002** includes fabricating at least integrated circuit **300D** or **500A** where a drain region and a source region of a transistor are fabricated in the substrate, and a first gate structure is fabricated. In these embodiments, the substrate is substrate **302a** which includes p-type dopants, the drain region is drain region **312b** or **512b** which includes n-type dopants, and the source region is source region **314b** or **514b** which includes n-type dopants, and the first gate structure is gate **330**.

[0413] In some embodiments, the formation of the source/drain features of at least transistor **360b** or **560b** in the substrate includes, removing a portion of the substrate to form recesses at an edge of each spacer **320a**, **320b**, and a filling process is then performed by filling the recesses in the substrate. In some embodiments, the recesses are etched, for example, a wet etching or a dry

etching, after removal of a pad oxide layer or a sacrificial oxide layer. In some embodiments, the etch process is performed to remove a top surface portion of the active region adjacent to an isolation region, such as STI region **308**, **309**, **310** or **311**. In some embodiments, the filling process is performed by an epitaxy or epitaxial (epi) process. In some embodiments, the recesses are filled using a growth process which is concurrent with an etch process where a growth rate of the growth process is greater than an etch rate of the etch process. In some embodiments, the recesses are filled using a combination of growth process and etch process. For example, a layer of material is grown in the recess and then the grown material is subjected to an etch process to remove a portion of the material. Then a subsequent growth process is performed on the etched material until a desired thickness of the material in the recess is achieved. In some embodiments, the growth process continues until a top surface of the material is above the top surface of the substrate. In some embodiments, the growth process is continued until the top surface of the material is co-planar with the top surface of the substrate.

[0414] In some embodiments, a portion of the substrate is removed by an isotropic or an anisotropic etch process. The etch process selectively etches the substrate without etching gate structure **230** and spacers **220**. In some embodiments, the etch process is performed using a reactive ion etch (RIE), wet etching, or other suitable techniques. In some embodiments, a semiconductor material is deposited in the recesses to form the source/drain features. In some embodiments, an epi process is performed to deposit the semiconductor material in the recesses. In some embodiments, the epi process includes a selective epitaxy growth (SEG) process, CVD process, molecular beam epitaxy (MBE), other suitable processes, and/or combination thereof. The epi process uses gaseous and/or liquid precursors, which interacts with a composition of substrate **302a** or **302b**. In some embodiments, the source/drain features include epitaxially grown silicon (epi Si), silicon carbide, or silicon germanium. Source/drain features of the IC device associated with gate structure **330** are in-situ doped or undoped during the epi process in some instances. When source/drain features are undoped during the epi process, source/drain features are doped during a subsequent process in some instances. The subsequent doping process is achieved by an ion implantation, plasma immersion ion implantation, gas and/or solid source diffusion, other suitable processes, and/or combination thereof. In some embodiments, source/drain features are further exposed to annealing processes after forming source/drain features and/or after the subsequent doping process.

[0415] In some embodiments, at least fabricating the gate regions of operation **1002**, **1004** or **1006** (described below) includes at least fabricating gate structure **330**. In some embodiments, at least fabricating the gate regions of operation **1002**, **1004** or **1006** (described below) includes performing one or more deposition processes to form one or more dielectric material layers. In some embodiments, a deposition process includes a chemical vapor deposition (CVD), a plasma enhanced CVD (PECVD), an atomic layer deposition (ALD), or other process suitable for depositing one or more material layers. In some embodiments, fabricating the gate regions includes performing one or more deposition processes to form one or more conductive material layers. In some embodiments, fabricating the gate regions includes forming gate electrodes or dummy gate electrodes. In some embodiments, fabricating the gate regions includes depositing or growing at least one dielectric layer, e.g., gate dielectric. In some embodiments, gate regions are formed using a doped or non-doped polycrystalline silicon (or polysilicon). In some embodiments, the gate regions include a metal, such as Al, Cu, W, Ti, Ta, TiN, TaN, NiSi, CoSi, other suitable conductive materials, or combinations thereof. Other materials for the gate regions are within the scope of the present disclosure.

[0416] In some embodiments, operation **1002** includes fabricating integrated circuit **300B** where P-type region **352a** and N-type region **356a** are formed in substrate **302a**, and the formation of P-type region **352a** and N-type region **356a** are similar to the formation of source/drain regions of at least integrated circuit **300A**, **300D**, **500A** or **500D**, and similar detailed description is therefore omitted.

[0417] In some embodiments, operation **1002** includes fabricating integrated circuit **500B** where N-type region **552a** and P-type region **556a** are formed in substrate **302a**, and the formation of N-type region **552a** and P-type region **556a** are similar to the formation of source/drain regions of at least integrated circuit **300A**, **300D**, **500A** or **500D**, and similar detailed description is therefore omitted.

[0418] In some embodiments, operation **1002** includes fabricating integrated circuit **300C** or **500E** where N-type region **366a** is formed in substrate **302a**, P-type region **362a** is formed in N-type region **366a**, and the formation of P-type region **362a** and N-type region **366a** are similar to the formation of source/drain regions of at least integrated circuit **300A**, **300D**, **500A** or **500D**, and similar detailed description is therefore omitted.

[0419] In some embodiments, operation **1002** includes fabricating integrated circuit **300E** or **500C** where P-type region **366b** is formed in substrate **302b**, N-type region **362b** is formed in P-type region **366b**, and the formation of N-type region **362b** and P-type region **366b** are similar to the formation of source/drain regions of at least integrated circuit **300A**, **300D**, **500A** or **500D**, and similar detailed description is therefore omitted.

[0420] In some embodiments, the regions with p-type dopants of sources/drains, tap regions, p-type region have a dopant concentration ranging from 1×10^{12} atoms/cm³ to 1×10^{14} atoms/cm³. Other values or ranges of dopant concentrations for p-type dopants are within the scope of the present disclosure. In some embodiments, the regions with n-type dopants of sources/drains, tap regions, n-type region have a dopant concentration ranging from 1×10^{12} atoms/cm³ to 1×10^{14} atoms/cm³. Other values or ranges of dopant concentrations for n-type dopants are within the scope of the present disclosure.

[0421] In some embodiments, at least operation **1002**, **1004** or **1006** further includes fabricating a signal tap region on the front-side of the wafer. In some embodiments, the signal tap region of method **1000** includes at least tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b**.

[0422] In some embodiments, at least tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** is a heavily doped p-region. In some embodiments, at least tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** is a heavily doped n-region.

[0423] In some embodiments, at least tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** is formed by ion implantation. The power of the ion implantation ranges from about 1500 k electron volts (eV) to about 8000 k eV. Other values or ranges of power for ion implantation are within the scope of the present disclosure.

[0424] In some embodiments, tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** is epitaxially grown. In some embodiments, tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** comprises an epi-layer grown over substrate **302a** or **302b**. In some embodiments, the epi-layer is doped by adding dopants during the epitaxial process. In some embodiments, the epi-layer is doped by ion implantation after the epi-layer is formed. In some embodiments, tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** is formed by doping substrate **302a** or **302b**. In some embodiments, the doping is performed by ion implantation. In some embodiments, tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** has a dopant concentration ranging from 1×10^{12} atoms/cm³ to 1×10^{14} atoms/cm³. Other values or ranges of dopant concentrations for tap region **316a**, **316b**, **317a**, **317b**, **516a**, **516b**, **517a** or **517b** are within the scope of the present disclosure.

[0425] In operation **1004** of method **1000**, a discharging circuit is fabricated on the wafer. In some embodiments, the discharging circuit of method **1000** includes at least discharging circuit **210** or **410**. In some embodiments, the discharging circuit of method **1000** includes at least NMOS transistor **N3** or PMOS transistor **P2**.

[0426] In some embodiments, operation **1004** includes fabricating a second well in a substrate, fabricating a second source region in the second well, fabricating a second drain region in the second well, and fabricating a second gate structure. In some embodiments, operation **1004** includes fabricating the second source region in the substrate, fabricating the second drain region in

the substrate, and fabricating the second gate structure.

[0427] In some embodiments, at least fabricating the second source/drain regions or the second gate structure of operation **1004** is similar to the formation of first source/drain regions or the first gate structure of operation **1002** (described above), and similar detailed description is omitted.

[0428] In operation **1006** of method **1000**, a detection circuit is fabricated on the wafer. In some embodiments, the detection circuit of method **1000** includes at least detection circuit **202** or **402**. In some embodiments, the detection circuit of method **1000** includes at least capacitor **C1** or **C2**, resistor **R1** or **R2**, NMOS transistor **N1** or PMOS transistor **P1**.

[0429] In some embodiments, operation **1006** includes fabricating a third well in a substrate, fabricating a third source region in the third well, fabricating a third drain region in the third well, and fabricating a third gate structure. In some embodiments, operation **1006** includes fabricating the third source region in the substrate, fabricating the third drain region in the substrate, and fabricating the third gate structure.

[0430] In some embodiments, at least fabricating the third source/drain regions or the third gate structure of operation **1006** is similar to the formation of first source/drain regions or the first gate structure of operation **1002** (described above), and similar detailed description is omitted.

[0431] In some embodiments, capacitor **C1** or **C2** of method **1000** is a transistor-coupled capacitor, and fabricating the transistor-coupled capacitor is similar to fabricating the NMOS transistor **N1** or PMOS transistor **P1**, but the drain, source and body are coupled together, and similar detailed description is omitted. In some embodiments, operation **1006** is repeated to form NMOS transistor **N1**, PMOS transistor **P1** and at least capacitor **C1** or **C2**.

[0432] In operation **1008** of method **1000**, a set of conductive structures is fabricated on a front-side of the wafer. In some embodiments, operation **1008** includes depositing the set of conductive structures on the front-side of the wafer. In some embodiments, the set of conductive structures of method **1000** includes at least conductive structure **390**. In some embodiments, the set of conductive structures of method **1000** includes conductive structures (not shown) configured to connect one or more of the gate, the drain, the source, the body, node **Nd1**, node **Nd2** or node **Nd3** of FIG. 3A-3B, 3D, 5A-5B or 5D, are similar to conductive structure **390**, and similar detailed description is omitted.

[0433] In some embodiments, the set of conductive structures of method **1000** are formed using a combination of photolithography and material removal processes to form openings in an insulating layer (not shown) over the substrate. In some embodiments, the photolithography process includes patterning a photoresist, such as a positive photoresist or a negative photoresist. In some embodiments, the photolithography process includes forming a hard mask, an antireflective structure, or another suitable photolithography structure. In some embodiments, the material removal process includes a wet etching process, a dry etching process, an RIE process, laser drilling or another suitable etching process. The openings are then filled with conductive material, e.g., copper, aluminum, titanium, nickel, tungsten, or other suitable conductive material. In some embodiments, the openings are filled using CVD, PVD, sputtering, ALD or other suitable formation process.

[0434] In some embodiments, one or more of the operations of method **1000** is not performed. In some embodiments, one or more of the operations of method **1000** is repeated. In some embodiments, method **1000** is repeated.

[0435] Other diode types or numbers of diodes, or transistor types or other numbers of transistors in at least integrated circuit **100**, **200A-200E**, **300A-300E**, **400A-400E**, **500A-500E**, **600A-600B**, **700A-700B** and **800A-800B** of corresponding FIGS. 1, 2A-2E, 3A-3E, 4A-4E, 5A-5E, 6A-6B, 7A-7B and 8A-8B are within the scope of the present disclosure.

[0436] Furthermore, various NMOS or PMOS transistors shown in FIGS. 2A-2E and 4A-4E are of a particular dopant type (e.g., N-type or P-type) and are for illustration purposes. Embodiments of the disclosure are not limited to a particular transistor type, and one or more of the PMOS or

NMOS transistors shown in FIGS. 2A-2E and 4A-4E can be substituted with a corresponding transistor of a different transistor/dopant type. Similarly, the low or high logical value of various signals used in the above description is also used for illustration. Embodiments of the disclosure are not limited to a particular logical value when a signal is activated and/or deactivated. Selecting different logical values is within the scope of various embodiments. Selecting different numbers of NMOS/PMOS transistors in 2A-2E and 4A-4E is within the scope of various embodiments.

[0437] One aspect of this description relates to an electrostatic discharge (ESD) circuit. In some embodiments, the ESD circuit includes a first ESD detection circuit coupled between a first node and a second node, the first node having a first voltage and the second node having a second voltage. In some embodiments, the ESD circuit further includes a first clamp circuit comprising a first transistor of a first type, the first transistor coupled between the first node, the second node and a third node. In some embodiments, the ESD circuit further includes a first ESD assist circuit coupled between the second node and the third node, and configured to clamp a third voltage of the third node at the second voltage during an ESD event at the first node or the second node. In some embodiments, the ESD circuit further includes a second ESD detection circuit coupled between the second node and a fourth node. In some embodiments, the ESD circuit further includes a second clamp circuit comprising a second transistor of the first type, the second transistor coupled between the second node, the fourth node and a fifth node. In some embodiments, the ESD circuit further includes a second ESD assist circuit coupled between the fourth node and the fifth node, and configured to clamp a fourth voltage of the fifth node at the second voltage during an ESD event at the second node or the fourth node.

[0438] Another aspect of this description relates to an ESD circuit. In some embodiments, the ESD circuit includes a first diode coupled between a first node and an input output (IO) pad. In some embodiments, the ESD circuit further includes a second diode coupled between the IO pad and a second node. In some embodiments, the ESD circuit further includes an ESD discharging circuit between the first node and the second node. In some embodiments, the ESD discharging circuit includes a first ESD detection circuit coupled between the first node and a third node, the first node having a first voltage and the third node having a second voltage. In some embodiments, the ESD discharging circuit further includes a first clamp circuit comprising a first transistor of a first type, the first transistor coupled to the first node, the third node and a fourth node. In some embodiments, the ESD discharging circuit further includes a first ESD assist circuit coupled between the third node and the fourth node, and configured to clamp a third voltage of the fourth node at the first voltage during an ESD event at the first node or the second node. In some embodiments, the ESD discharging circuit further includes a second ESD detection circuit coupled between the second node and the third node. In some embodiments, the ESD discharging circuit further includes a second clamp circuit comprising a second transistor of the first type, the second transistor coupled to the second node, the third node and a fifth node. In some embodiments, the ESD discharging circuit further includes a second ESD assist circuit coupled between the second node and the fifth node, and configured to clamp a fourth voltage of the fifth node at the second voltage during an ESD event at the third node.

[0439] Yet another aspect of this description relates to a method of operating an ESD circuit. In some embodiments, the method includes detecting, by a detection circuit, a first ESD event at a first node thereby causing the detection circuit to charge a first gate of a first transistor of a first clamp circuit, the first ESD event including a first ESD voltage of the first node being greater than a reference supply voltage of a reference voltage supply. In some embodiments, the method further includes turning on a first ESD assist circuit in response to the first ESD event at the first node thereby causing the first ESD assist circuit to clamp a first voltage of the first gate of the first transistor of the first clamp circuit at a second voltage of a second node, the first clamp circuit being coupled between the first node and the second node, and the first ESD assist circuit being coupled between at least the first node and a third node, and the first ESD assist circuit including a

first body diode. In some embodiments, the method further includes turning on the first clamp circuit in response to the first ESD event and the first ESD assist circuit clamping the first voltage of the first gate at the second voltage. In some embodiments, the method further includes discharging a first ESD current of the first ESD event in a first ESD direction from the first node to the second node by the first transistor. In some embodiments, the method further includes turning on a second ESD assist circuit in response to the first ESD current of the first ESD event at the second node thereby causing the second ESD assist circuit to clamp a third voltage of a second gate of a second transistor of a second clamp circuit at a fourth voltage of a fourth node, the second clamp circuit being coupled between the second node and the fourth node, and the second ESD assist circuit being coupled between at least the fourth node and a fifth node, and the second ESD assist circuit including a second body diode. In some embodiments, the method further includes turning on the second clamp circuit in response to the first ESD current at the second node and the second ESD assist circuit clamping the third voltage of the second gate at the fourth voltage. In some embodiments, the method further includes discharging the first ESD current of the first ESD event in the first ESD direction from the second node to the fourth node by the second transistor.

[0440] A number of embodiments have been described. It will nevertheless be understood that various modifications may be made without departing from the spirit and scope of the disclosure. For example, various transistors being shown as a particular dopant type (e.g., N-type or P-type Metal Oxide Semiconductor (NMOS or PMOS)) are for illustration purposes. Embodiments of the disclosure are not limited to a particular type. Selecting different dopant types for a particular transistor is within the scope of various embodiments. The low or high logical value of various signals used in the above description is also for illustration. Various embodiments are not limited to a particular logical value when a signal is activated and/or deactivated. Selecting different logical values is within the scope of various embodiments. In various embodiments, a transistor functions as a switch. A switching circuit used in place of a transistor is within the scope of various embodiments. In various embodiments, a source of a transistor can be configured as a drain, and a drain can be configured as a source. As such, the term source and drain are used interchangeably. Various signals are generated by corresponding circuits, but, for simplicity, the circuits are not shown.

[0441] Various figures show capacitive circuits using discrete capacitors for illustration. Equivalent circuitry may be used. For example, a capacitive device, circuitry or network (e.g., a combination of capacitors, capacitive elements, devices, circuitry, or the like) can be used in place of the discrete capacitor. The above illustrations include exemplary steps, but the steps are not necessarily performed in the order shown. Steps may be added, replaced, changed order, and/or eliminated as appropriate, in accordance with the spirit and scope of disclosed embodiments.

[0442] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An electrostatic discharge (ESD) circuit comprising: a first ESD detection circuit coupled between a first node and a second node, the first node having a first voltage and the second node having a second voltage; a first clamp circuit comprising a first transistor of a first type, the first transistor coupled between the first node, the second node and a third node; a first ESD assist

circuit coupled between the second node and the third node, and configured to clamp a third voltage of the third node at the second voltage during an ESD event at the first node or the second node; a second ESD detection circuit coupled between the second node and a fourth node; a second clamp circuit comprising a second transistor of the first type, the second transistor coupled between the second node, the fourth node and a fifth node; and a second ESD assist circuit coupled between the fourth node and the fifth node, and configured to clamp a fourth voltage of the fifth node at the second voltage during an ESD event at the second node or the fourth node.

2. The ESD circuit of claim 1, further comprising: a first conductive line extending in a first direction, and configured to supply the first voltage; a second conductive line extending in the first direction, and configured to supply the second voltage; a third conductive line extending in the first direction, and the third conductive line including the third node; a fourth conductive line extending in the first direction, and configured to supply the first voltage; and a fifth conductive line extending in the first direction, and configured to supply the second voltage.

3. The ESD circuit of claim 2, wherein the first node is part of the first conductive line or the fourth conductive line; the second node is part of the second conductive line or the fifth conductive line; each of the first conductive line, the second conductive line, the third conductive line, the fourth conductive line and the fifth conductive line are separated from each other in a second direction different from the first direction; the third conductive line is between the first conductive line and the fourth conductive line; the first conductive line is between the third conductive line and the second conductive line; and the fourth conductive line is between the third conductive line and the fifth conductive line.

4. The ESD circuit of claim 1, wherein the first ESD assist circuit comprises: a diode coupled between the second node and the third node, the diode having an anode coupled to the third node and the first ESD detection circuit, and a cathode coupled to the second node.

5. The ESD circuit of claim 1, wherein the first ESD assist circuit comprises: a third transistor of the first type, and having a gate of the third transistor, a drain of the third transistor, a source of the third transistor, and a body terminal of the third transistor, the drain of the third transistor being coupled to the third node and the first ESD detection circuit, and each of the second node, the gate of the third transistor, the source of the third transistor and the body terminal of the third transistor being coupled together.

6. The ESD circuit of claim 5, wherein the third transistor comprises: a first active region in a substrate, the first active region corresponding to the drain of the third transistor; a second active region in the substrate, the second active region corresponding to the source of the third transistor; and a third active region in the substrate, the third active region corresponding to the body terminal of the third transistor, wherein the second active region and the third active region are coupled together, and the first active region and the third active region are configured as a body diode of the third transistor.

7. The ESD circuit of claim 1, wherein the first ESD assist circuit comprises: a first bipolar junction transistor having a base of the first bipolar junction transistor, a collector of the first bipolar junction transistor and an emitter of the first bipolar junction transistor, the collector of the first bipolar junction transistor being coupled to the third node and the first ESD detection circuit, and each of the first node, the base of the first bipolar junction transistor and the emitter of the first bipolar junction transistor being coupled together.

8. The ESD circuit of claim 1, wherein the first ESD assist circuit comprises: a third transistor of a second type different from the first type, and having a gate of the third transistor, a drain of the third transistor, a source of the third transistor, and a body terminal of the third transistor, the drain of the third transistor being coupled to the second node, and each of the gate of the third transistor, the source of the third transistor, the body terminal of the third transistor, the third node and the first ESD detection circuit being coupled together.

9. The ESD circuit of claim 8, wherein the third transistor comprises: a first active region in a

substrate, the first active region corresponding to the drain of the third transistor; a second active region in the substrate, the second active region corresponding to the source of the third transistor; and a third active region in the substrate, the third active region corresponding to the body terminal of the third transistor, wherein the second active region and the third active region are coupled together, and the first active region and the third active region are configured as a body diode of the third transistor.

10. The ESD circuit of claim 1, wherein the first ESD assist circuit comprises: a first bipolar junction transistor having a base of the first bipolar junction transistor, a collector of the first bipolar junction transistor and an emitter of the first bipolar junction transistor, the collector of the first bipolar junction transistor being coupled to the second node, and each of the emitter of the first bipolar junction transistor, the base of the first bipolar junction transistor, the third node and the first ESD detection circuit being coupled together.

11. An electrostatic discharge (ESD) circuit comprising: a first diode coupled between a first node and an input output (IO) pad; a second diode coupled between the IO pad and a second node; and an ESD discharging circuit between the first node and the second node, the ESD discharging circuit comprising: a first ESD detection circuit coupled between the first node and a third node, the first node having a first voltage and the third node having a second voltage; a first clamp circuit comprising a first transistor of a first type, the first transistor coupled to the first node, the third node and a fourth node; a first ESD assist circuit coupled between the third node and the fourth node, and configured to clamp a third voltage of the fourth node at the first voltage during an ESD event at the first node or the second node; a second ESD detection circuit coupled between the second node and the third node; a second clamp circuit comprising a second transistor of the first type, the second transistor coupled to the second node, the third node and a fifth node; and a second ESD assist circuit coupled between the second node and the fifth node, and configured to clamp a fourth voltage of the fifth node at the second voltage during an ESD event at the third node.

12. The ESD circuit of claim 11, wherein the first ESD assist circuit comprises: a diode coupled between the fourth node and the third node, the diode having an anode coupled to the fourth node and the first ESD detection circuit, and a cathode coupled to the third node.

13. The ESD circuit of claim 11, wherein the first ESD assist circuit comprises: a first bipolar junction transistor having a base of the first bipolar junction transistor, a collector of the first bipolar junction transistor and an emitter of the first bipolar junction transistor, the collector of the first bipolar junction transistor being coupled to the fourth node and the first ESD detection circuit, and each of the third node, the base of the first bipolar junction transistor and the emitter of the first bipolar junction transistor being coupled together.

14. The ESD circuit of claim 11, wherein the first ESD assist circuit comprises: a third transistor of the first type, and having a gate of the third transistor, a drain of the third transistor, a source of the third transistor, and a body terminal of the third transistor, the drain of the third transistor being coupled to the fourth node and the first ESD detection circuit, and each of the third node, the gate of the third transistor, the source of the third transistor and the body terminal of the third transistor being coupled together.

15. The ESD circuit of claim 11, further comprising: a first conductive line extending in a first direction, and configured to supply the first voltage; a second conductive line extending in the first direction, and configured to supply the second voltage; a third conductive line extending in the first direction, and the third conductive line including the third node; a fourth conductive line extending in the first direction, and configured to supply the first voltage; and a fifth conductive line extending in the first direction, and configured to supply the second voltage.

16. The ESD circuit of claim 11, wherein the first ESD detection circuit comprises: a first resistor coupled between the first node and a first input node; a first capacitor coupled between the first input node and the third node; and a first inverter coupled to the first node, the fourth node, the third node, the first input node and the first ESD assist circuit.

17. The ESD circuit of claim 16, wherein the first inverter comprises: a third transistor of the first type, the third transistor having a gate of the third transistor, a drain of the third transistor, a source of the third transistor, and a body terminal of the third transistor; and a fourth transistor of a second type different from the first type, the fourth transistor having a gate of the fourth transistor, a drain of the fourth transistor, a source of the fourth transistor, and a body terminal of the fourth transistor; wherein each of the gate of the third transistor, the gate of the fourth transistor, and the first input node are coupled together; each of the source of the third transistor, the body terminal of the third transistor and the third node are coupled together; each of the source of the fourth transistor, the body terminal of the fourth transistor and the first node are coupled together; and each of the drain of the third transistor, the drain of the fourth transistor and the third node are coupled together.

18. The ESD circuit of claim 17, wherein the second ESD detection circuit comprises: a second resistor coupled between the third node and a second input node; a second capacitor coupled between the second input node and the second node; and a second inverter coupled to the second node, the third node, the fifth node, the second input node and the second ESD assist circuit.

19. The ESD circuit of claim 18, wherein the second inverter comprises: a fifth transistor of the first type, the fifth transistor having a gate of the fifth transistor, a drain of the fifth transistor, a source of the fifth transistor, and a body terminal of the fifth transistor; and a sixth transistor of the second type, the sixth transistor having a gate of the sixth transistor, a drain of the sixth transistor, a source of the sixth transistor, and a body terminal of the sixth transistor; wherein each of the gate of the fifth transistor, the gate of the sixth transistor, and the second input node are coupled together; each of the source of the fifth transistor, the body terminal of the fifth transistor and the second node are coupled together; each of the source of the sixth transistor, the body terminal of the sixth transistor and the third node are coupled together; and each of the drain of the fifth transistor, the drain of the sixth transistor and the fifth node are coupled together.

20. A method of operating an electrostatic discharge (ESD) circuit, the method comprising: detecting, by a detection circuit, a first ESD event at a first node thereby causing the detection circuit to charge a first gate of a first transistor of a first clamp circuit, the first ESD event including a first ESD voltage of the first node being greater than a reference supply voltage of a reference voltage supply; turning on a first ESD assist circuit in response to the first ESD event at the first node thereby causing the first ESD assist circuit to clamp a first voltage of the first gate of the first transistor of the first clamp circuit at a second voltage of a second node, the first clamp circuit being coupled between the first node and the second node, and the first ESD assist circuit being coupled between at least the first node and a third node, and the first ESD assist circuit including a first body diode; turning on the first clamp circuit in response to the first ESD event and the first ESD assist circuit clamping the first voltage of the first gate at the second voltage; discharging a first ESD current of the first ESD event in a first ESD direction from the first node to the second node by the first transistor; turning on a second ESD assist circuit in response to the first ESD current of the first ESD event at the second node thereby causing the second ESD assist circuit to clamp a third voltage of a second gate of a second transistor of a second clamp circuit at a fourth voltage of a fourth node, the second clamp circuit being coupled between the second node and the fourth node, and the second ESD assist circuit being coupled between at least the fourth node and a fifth node, and the second ESD assist circuit including a second body diode; turning on the second clamp circuit in response to the first ESD current at the second node and the second ESD assist circuit clamping the third voltage of the second gate at the fourth voltage; and discharging the first ESD current of the first ESD event in the first ESD direction from the second node to the fourth node by the second transistor.
